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MapLight Therapeutics | North America

MapLight's ML007-C-MA: A Game Changer in Antipsychotics? Initiate at Overweight, \$34 price target

We initiate coverage of MapLight Therapeutics with an Overweight rating and a \$34 PT. We see MPLT's lead asset, ML007-C-MA as having best in class potential in a >\$20bn market for antipsychotics.

There is a gap in the current antipsychotic landscape... Approved antipsychotics are all very similar, with high switch rates, positioning novel drug mechanisms to gain market share. Although there are dozens of approved antipsychotics, 30% of US schizophrenia patients show no response, and another 30-45% are poorly controlled. Muscarinic receptor agonism is the first new therapeutic strategy in decades for lowering psychosis, and has been clinically validated in schizophrenia by Cobenyl (BMY's muscarinic agonist). MPLT's lead asset, ML-007C-MA, uses learnings from Cobenyl and others, and could be a best-in-class muscarinic receptor agonist.

...which ML007-C-MA is well positioned to fill. Psychosis is common in schizophrenia (2-3M US adults) and treatment is a >\$20bn market by 2030. For the 6-8M US adults with Alzheimer's Disease, up to 40% have a psychotic episode (Alzheimer's Disease Psychosis or ADP). There are no FDA approved therapies for ADP, but reports suggest a third of AD patients have been prescribed an off-label antipsychotic. ML-007C-MA has precision-matched pharmacokinetics for greater activation of M1/M4 vs. Cobenyl, and lower rates of pro and anticholinergic side effects making it a potential best-in-class asset in this large market.

There are several catalysts ahead for de-risking ML007 (and muscarinic indication expansion). ML007-C-MA is in ongoing Phase 2 development for schizophrenia (Phase 2 ZEPHYR) and for AD Psychosis (Phase 2 VISTA), with topline results expected in 2H26 and 2H27, respectively. Ongoing Phase 3 ADEPT trials with COBENFY Phase 3 could de-risk expansion of muscarinics for AD Psychosis, with the first of four trials, ADEPT-2, expected to read out in 4Q25.

Valuation: Overweight with \$34 PT by DCF. We forecast \$1.4B in ML-007-C-MA revenues (\$850M from schizophrenia, 60% probability of success, plus \$590M for AD Psychosis, 50% probability of success), and \$230M in ML004 revenues in Autism Spectrum disorder (30% probability of success), for ~\$1.7B in peak risk-adjusted revenues. In our view, these forecasts may prove conservative given our "007" patient numbers ~8 years post launch are ~1/3 of Cobenyl's.

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MapLight Therapeutics (MPLT.O, MPLT US)

Biotechnology United States of America				
Stock Rating	Overweight			
Industry View	Attractive			
Price target	\$34.00			
Shr price, close (Nov 19, 2025)	\$14.60			
Mkt cap, curr (mm)	\$717			
52-Week Range	\$20.86-13.33			
Fiscal Year Ending	12/23	12/24e	12/25e	12/26e
EPS (\$)**	(5.03)	(6.27)	(5.49)	(3.76)
Prior EPS (\$)**	-	-	-	-
P/E	NM	NM	NM	NM
EPS (\$)§	-	-	-	-
Div yld (%)	-	-	-	-

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

QUARTERLY EPS (\$)					
Quarter	2023	2024e Prior	2024e Current	2025e Prior	2025e Current
Q1	-	(1.68)a	-	(1.75)	-
Q2	-	(1.51)	-	(1.14)	-
Q3	-	(1.60)	-	(1.65)	-
Q4	-	(1.49)	-	(1.25)	-

e = Morgan Stanley Research estimates, a = Actual Company reported data

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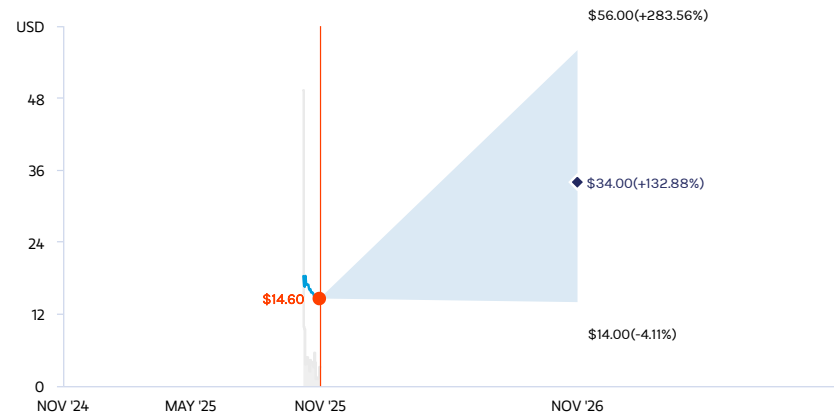
Risk Reward – MapLight Therapeutics (MPLT.O)

ML-007C-MA in Schizophrenia (and in Alzheimer's disease) Drives the Risk/Reward

PRICE TARGET \$34.00

Our PT is derived from a discounted cash flow analysis that assumes an 15% WACC, a stock beta of 2.0, and 2.5% terminal growth.

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

ML-007C-MA has potential to be a best-in-class muscarinic receptor agonist. ML-007 has precision-matched pharmacokinetics for greater activation of M1/M4 vs. Cobenfy (and reduced side effects).

Risk Reward Themes

Disruption: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	\$56.00	BASE CASE	\$34.00	BEAR CASE	\$14.00
Bull Case- Clinical Derisking		Current Base Case		Bear Case- Poor Clinical Data	
\$2B in risk adjust revenue. \$1B in schizophrenia (70% PoS), \$700M for ADP (50% PoS), \$250M for ML004 in Autism (40% PoS).		\$1.6B in risk adjusted revenues. \$850M for schizophrenia (60% PoS), \$600M for ADP (50% PoS), \$250M for ML004 in Autism (30% PoS).		\$1.3B in risk adjusted revenues. We ascribe lower PoS for ML007 in schizophrenia (50%), in ADP (40%), and for ML004 in Autism (20%).	
<i>Market shares fixed across base and bull case at 20%,5%,3%, respectively.</i>		<i>Market shares fixed across base and bull case at 20%,5%,3%, respectively.</i>		<i>Market shares fixed across base and bull case at 20%,5%,3%, respectively.</i>	

Risk Reward – MapLight Therapeutics (MPLT.O)

KEY EARNINGS INPUTS

Drivers	2023	2024e	2025e	2026e
Total Revenues (\$, mm)	0	0	0	0
FCF (\$, mm)	(54)	(81)	(142)	(189)
Peak Revenues (\$, mm)	1,394	1,394	1,394	1,394

INVESTMENT DRIVERS

- **ML007-C-MA** Phase 2 ZEPHYR for schizophrenia (2H26)
- **ML007-C-MA** Phase 2 VISTA for Alzheimer's disease psychosis (2H27)
- **Cobenfy** Phase 3 ADEPT studies (4Q25, 2026)

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

RISKS TO PT/RATING

RISKS TO UPSIDE

- Positive Cobenfy Phase 3 ADEPT results (BMY)
- Cobenfy commercial performance (BMY)

RISKS TO DOWNSIDE

- Novel MOAs in Schizophrenia (LBRX).
- Competitive muscarinic pipeline expansion (NBIX)

MS ESTIMATES VS. CONSENSUS

FY 2024e

Sales / Revenue (\$, mm) ◆ 0
Note: There are not sufficient brokers supplying consensus data for this metric

Net income (\$, mm) ◆ (78)
Note: There are not sufficient brokers supplying consensus data for this metric

EPS (\$) ◆ (6.27)
Note: There are not sufficient brokers supplying consensus data for this metric

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Investment Summary

We are initiating coverage of MapLight Therapeutics (MPLT) with an Overweight rating and a \$34 price target. MapLight is a clinical stage company developing therapies to treat psychiatric symptoms in highly prevalent CNS disorders, like schizophrenia and Alzheimer's disease. The company is led by a team with a deep understanding of neural circuit science, including seminal founders in the fields of optogenetics and whole-tissue volumetric imaging in dopamine/serotonin brain circuitry. MapLight's integrated STARmap discovery platform can identify unique drug-able targets (in distinct genetically/anatomically defined circuits) to alleviate symptoms in CNS disorders.

The company's lead asset is ML007-C-MA, a M1/M4 muscarinic agonist combined with fesoterodine, a peripheral anticholinergic (PAC). ML007-C-MA is in clinical development for schizophrenia (Phase 2 ZEPHYR) and for Alzheimer's Disease Psychosis (Phase 2 VISTA), with topline results expected in 2H26 and 2H27, respectively. MapLight is also developing ML-004, a 5-HT1B/1D agonist in Phase 2 for Autism Spectrum Disorder, as well as earlier-stage preclinical programs. We see ML007-C-MA as having best-in-class potential and generating \$2.6bn in revenue by 2035.

MapLight's development pipeline includes other Phase 2 programs and earlier stage preclinical programs. ML-004 (5-HT1B/1D agonist) is in ongoing development for Autism Spectrum Disorder (ASD) as part of Phase 2 IRIS. Early stage programs include ML-021 (preclinical M4-Antagonist for treatment of Motor Symptoms of Parkinson's Disease), expected to complete IND-enabling studies in 2H26, and ML-009 (preclinical GPR52 agonist for treatment of Hyperactivity, Agitation and Impulsivity, with expected preclinical candidate nomination in 2026).

Conservative assumptions support our \$34 DCF-based price target: We forecast out to Yr-8 post launch for ML-007C-MA. Here (2035) we expect ML-007C-MA revenue at ~53% of Cobenfy forecast Yr-8 post launch. We see ~\$2.6bn revenue in 2035 for ML007-C-MA. We do model a higher revenue/patient p.a. than Cobenfy as we expect that a better safety and tolerability profile will lead to greater patient compliance overall.

Key Debates Discussed in this Report

Debate 1: Muscarinics in schizophrenia: How does ML007 become best-in-class?

Debate 2: Muscarinics in Alzheimer's: Is there a coming paradigm shift?

Debate 3: Can ML004 help address the growing burden of Autism?

Exhibit 1: Summary of peak sales projections

Program	ML007-C-MA	ML007-C-MA	ML004
Indication	Schizophrenia	Alzheimer's Disease Psychosis	Autism Spectrum Disorder
US Addressable Patients, at Launch	501,300	1,701,300	2,414,000
Launch Year	2029	2030	2031
Annual Gross Price, at Launch	\$24,000	\$24,720	\$20,000
Probability of Success	60%	50%	30%
Peak Market Share	20%	5%	3%
Peak US Sales \$M (Risk-Adjusted)	\$1,423	\$1,178	\$780
Peak US Sales \$M (Unadjusted)	\$854	\$589	\$234

Source: Morgan Stanley Research

Key Investment Points

ML007-C-MA in schizophrenia is largely de-risked. Results with Cobenfy in Phase 2 EMERGENT-1 (and Phase 3 EMERGENT-2 and -3) de-risk the development of muscarinics for the treatment of psychosis in schizophrenia, but also suggest a narrow therapeutic index. Recall that for Cobenfy in schizophrenia, a PANSS total was -8.4 vs placebo at 5 weeks (EMERGENT-3), with similar directionality across EMERGENT-1/-2 including effects on positive and (to a lesser extent) negative symptoms. However, there have been several readouts in the space (positive results with Cobenfy in EMERGENT, negative emraclidine results in EMPOWER, and mixed perceptions of direclidine/NBI-568 Phase 2) which suggest efficacy is paramount to investors. ML007-C-MA has optimized PK between its agonist ML007 and PAC fesoterodine, which should help differentiate from Cobenfy's safety profile (while maintaining similar efficacy). We believe the setup leaves ML007-C-MA ideally positioned to both move the needle for patients, and alleviate investor concerns on the muscarinic agonist profile and prospects.

Expansion of muscarinic agonists in ADP would signal a paradigm shift in the treatment of psychosis. There is a growing need for the 6-8M US adults suffering from Alzheimer's disease, 40% of whom have psychotic events and a third of whom are on off-label antipsychotics. The ongoing Phase 3 ADEPT studies are expected to start to read out by the end of this year, and the results could de-risk expansion of muscarinics for treatment of psychosis in AD. Indeed, we believe Cobenfy ADEPT-2 topline results expected in 4Q25 have readthrough to MPLT and Phase 2 VISTA, and can provide a clear regulatory precedent.

ML004 can address a large unmet need recognized by regulators. In autism, there is strong regulatory desire to fast-track effective medications. Prevalence of Autism in the US has increased +500% over the last two decades. However, risperidone and aripiprazole are the only FDA-approved antipsychotics in ASD (for the treatment of irritability), in stark contrast to schizophrenia (where there are dozens of atypical antipsychotics approved). ML004 Phase 2 IRIS is ongoing. Considering precedent set by risperidone and aripiprazole, plus potential for similar pharmacology to 5-HT1B/1D agonist ML-004, we believe ML004 is being largely overlooked and can become a value driver in the not too distant future.

Key Upcoming Catalysts

Exhibit 2: Upcoming catalysts and milestones (most stock-impactful events highlighted)

Drug	Target	Indication	Milestone	Timing	Impact
ML-007C-MA	M1/M4 agonist	Schizophrenia	Phase 2 topline results	2H26	***
ML-007C-MA	M1/M4 agonist	Alzheimer's Disease Psychosis	Phase 2 topline results	2H27	***
ML-004	5-HT1B/1D agonist	Autism Spectrum Disorder (Sociability/Irritability)	Phase 2 ongoing	2H26	**
ML-021	M4 antagonist	Parkinson's Disease	IND-enabling studies	2H26	*
ML-009	GPR52 PAM	Hyperactivity/Impulsivity	Preclinical candidate nomination	2026	*

Source: Morgan Stanley Research, Company reports

Exhibit 3: MapLight clinical development pipeline

Program	Circuit	Indications	Preclinical	Phase 1	Phase 2	Phase 3	Anticipated Milestones
ML-007C-MA M ₁ /M ₄ agonist co-formulated with PAC	Direct and Indirect Pathways	Schizophrenia	ZEPHYR				Phase 2 ongoing
		Alzheimer's Disease Psychosis	VISTA				Phase 2 ongoing
ML-004 5-HT _{1B/1D} agonist	Dorsal Raphe to Nucleus Accumbens	Autism Spectrum Disorder Sociability/Irritability	IRIS				Phase 2 ongoing
ML-021 M ₄ antagonist	Direct Pathway	Parkinson's Disease					IND-enabling studies
ML-009 GPR52 PAM	Indirect Pathway	Hyperactivity/Impulsivity					To be disclosed

Potential in other indications being explored

Source: Company reports

Valuation

Exhibit 4: Discounted Cash Flow analysis (\$M except per share)

(\$M except per share)	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	Terminal
Operating income	\$mn	(31)	(57)	(83)	(166)	(205)	(220)	(327)	(360)	(197)	(28)	104	244	421	609	807
<i>growth</i>	%		83%	45%	100%	24%	8%	48%	10%	(45%)	(86%)	(466%)	135%	73%	44%	32%
Tax	\$mn	0	0	0	0	0	0	0	0	0	0	(5)	(11)	(19)	(26)	(108)
<i>rate</i>	%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	4.2%	4.2%	4.2%	4.2%	13.0%
Depreciation and amortization	\$mn	1	1	1	1	1	1	1	1	2	2	2	3	3	4	4
Capex	\$mn	-1	0	-1	-2	-3	-3	-5	-6	-6	-6	-7	-8	-8	-9	-10
<i>capex/depreciation</i>	x	130%	83%	110%	372%	381%	342%	418%	435%	323%	286%	285%	282%	265%	250%	251%
Working capital	\$mn	-2	-6	-6	-20	-25	-27	-42	-54	-31	-7	11	30	54	80	107
Change in working capital	\$mn	-2	-4	0	-14	-5	-2	-15	-12	23	23	18	19	24	26	27
Free Cash Flow (FCF)	\$mn	(29)	(53)	(83)	(154)	(202)	(221)	(315)	(353)	(224)	(56)	87	231	410	604	882
<i>Growth</i>			NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	166%	78%	47%	46%
<i>Discount year</i>		(3)	(2)	(1)	0	1	2	3	4	5	6	7	8	9	10	11
Present Value of FCF	\$mn	\$0	\$0	\$0	\$0	(\$175)	(\$167)	(\$207)	(\$202)	(\$111)	(\$24)	\$33	\$76	\$117	\$149	\$190

Current NPV Analysis	
Discount rate	15%
Discount year	2025
Terminal growth rate	2.5%
Net enterprise value	\$1,231
Net Cash	\$421
Market Value	\$1,652
Shares Outstanding	49.1
NPV (per share)	\$34
Share price	\$ 17.00
Upside/Downside	98%
Funding mix	
Debt	0 0%
Market Capitalization	835 100%
Total Capitalization	835

Source: Morgan Stanley Research

We assume ~\$1.7B in peak risk-adjusted revenues by 2036. We derive our price target from a discounted cash flow (DCF) analysis using a 15% discount rate.

- **Discount Rate:** We use a 15% discount rate for companies without randomized Phase 2 data.
- **Terminal Growth:** We model revenues through 2036 and 2.5% terminal growth.
- **Revenue:** We model ML007-C-MA revenues for treatment of psychosis in schizophrenia (60% probability of success) and Alzheimer's Disease (50% probability of success), and ML004 revenues in Autism Spectrum disorder (30% probability of success). We ascribe no pipeline credit for earlier stage programs, and incorporate minimal operating expenses associated to their development. Our assumptions on launch years and peak sales are shown in [Exhibit 1](#).

Economics:

- Composition of matter and use of ML-007 patent to treat schizophrenia and ADP is expected to expire in 2031-2032. There are four pending U.S. provisional applications to treat schizophrenia and ADP, expected to expire in 2045 and exclusive of patent term extension, directed to pharmaceutical compositions containing ML-007 and Fesoterodine.
- MapLight has \$260M in cash/cash equivalents as of July 2025 (inclusive of \$200M in proceeds raised from initial closing of Series D). In October, MapLight raised gross proceeds (from IPO and Concurrent Private Placement) of \$296.3 million.
- MapLight projects operating expenses of ~\$200M in FY26 and ~\$220M in FY27 and will likely need to raise additional capital post its initial offering to fund operations through ML-007C-MA commercialization and launch.

Key risks to our price target: Risks include clinical, regulatory, and commercial.

Debate 1: Muscarinics in schizophrenia: How does ML007 become best-in-class?

Overview: MapLight's ML007-C-MA is a muscarinic agonist in Phase 2 development for the treatment of psychosis in schizophrenia (Phase 2 ZEPHYR) and the treatment of psychosis in Alzheimer's disease (Phase 2 VISTA). Similar to Cobenfy (fixed-dose of M1/M4 agonist xanomeline + a peripheral anticholinergic trospium), ML007-C-MA is a fixed-dose of ML007 (a potent M1/M4 agonist) + a peripheral anticholinergic fesoterodine.

Street's take: Muscarinic receptor agonism is the first novel therapeutic strategy in decades for lowering psychosis, and has been clinically validated in schizophrenia by first-in-class COBENFY. Commercial prospects for ML007-C-MA are therefore benchmarked to Cobenfy's, annualizing to >\$170M in its first full year. Estimates imply rapid uptake (\$470M in FY26, \$980M in FY27, \$1.6B in FY28), but on clinical prospects, the Street remains somewhat wary following negative emraclidine results and mixed perceptions of direclidine (NBI-568) Phase 2.

Our take: Muscarinics have to balance a narrow therapeutic index, but solving this is needed to fully realize the commercial opportunity in schizophrenia (i.e., moving outside of the ~30% of refractory patients). Indeed, there is an inherent trade-off for maximizing reduction of psychosis while ensuring lower rates of cholinergic side effects, but MapLight's ML007-C-MA agonist is differentiated. Phase 1 data for ML007-C-MA show a reduction in both pro- and anticholinergic adverse events vs. Cobenfy, driven by a stable ratio of muscarinic-to-anticholinergic achieved in brain:periphery. This results in better safety and improved tolerability in Phase 1, enabling higher ML007 exposures in brain and potentially greater efficacy.

Primer on Muscarinics: The 3rd Generation of Antipsychotics

1. What is a muscarinic, and what does it do? A muscarinic agonist is a small molecule antipsychotic that binds to and activates (muscarinic) acetylcholine receptors. There are five receptor subtypes (M1–M5), and muscarinics lower psychosis by activating the M1 and M4 receptor in the brain. However, M receptors are also expressed outside the body, leading to off-target events and a debate in industry on the preferred targeting strategy.

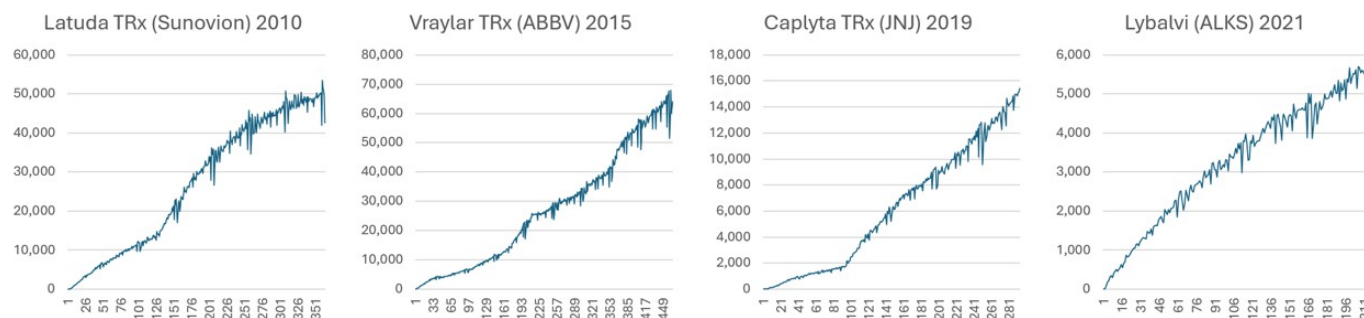
2. Muscarinic agonists are clinically validated and poised to disrupt a \$75B market. Antipsychotics are standard-of-care for lowering psychosis (hallucinations, delusions) in schizophrenia and other psychiatric disorders. There are dozens of FDA approved **first-generation (typical)** and **second-generation (atypical) antipsychotics** ([Exhibit 5](#)), most of which are approved in more than one indication (exception of Nuplazid). Annualizing to ~75M scripts (TRx) at an average net price of ~\$1,000/script/patient year, antipsychotics readily achieve blockbuster status (>\$1B) and the class generates >\$75B annually across indications, and we estimate ~20M TRx for the treatment of schizophrenia.

Exhibit 5: FDA approval history for typical, atypical and muscarinic antipsychotics

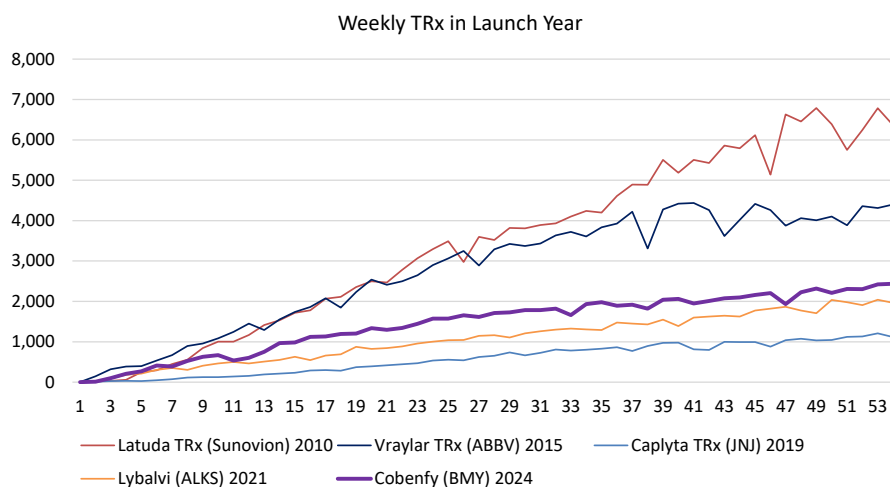
Drug	Approved Indication	Approval Year
Seroquel (quetiapine)	- Schizophrenia - Bipolar I disorder manic episodes - Bipolar disorder, depressive episodes	1997 2003 2006
Zyprexa (olanzapine)	- Schizophrenia - Bipolar I disorder manic or mixed episodes - Bipolar disorder, depressive episodes - Acute agitation in schizophrenia and bipolar I mania - Treatment Resistant Depression	1996 2000 (acute), 2004 (maintenance) 2003 2006 2009
Risperdal (risperidone)	- Schizophrenia - Bipolar mania - Irritability associated with autistic disorder - Bipolar disorder	1993 2003 2006 2025
Abilify (aripiprazole)	- Schizophrenia - Bipolar I disorder manic or mixed episodes - Agitation associated with schizophrenia or bipolar mania - Major depressive disorder - Irritability associated with autistic disorder - Tourette's disorder	2002 2004 (acute), 2005 (maintenance) 2007 2007 2009 2014
Invega (paliperidone)	- Schizophrenia - Schizoaffective disorder	2006 2014
Latuda (lurasidone)	- Schizophrenia - Bipolar depression	2010 2013
Vraylar (cariprazine)	- Schizophrenia - Bipolar disorder - Bipolar depression - Major depressive disorder	2015 (acute), 2017 (maintenance) 2015 2019 2022
Rexulti (brexpiprazole)	- Schizophrenia - Major depressive disorder - Treatment of Agitation Associated with Dementia Due to AD	2015 2015 2023
Nuplazid (pimavanserin)	Hallucinations and delusions associated with PDP	2016
Caplyta (lumateperone)	- Schizophrenia - Depressive episodes associated with bipolar I or II disorder - Major depressive disorder	2019 2021 2025
Lybalvi (olanzapine and samidorphan)	- Schizophrenia - Bipolar I disorder manic or mixed episodes	2021 2021 (acute), 2021 (maintenance)
Cobenfy (Xanomeline and trospium)	Schizophrenia	2024

Source: Morgan Stanley Research, FDA.gov

3. Schizophrenia launches are long and linear. Dozens of approved therapies ([Exhibit 5](#)) have led to a fragmented market with high switch rates, and as a result, launches for new molecules in schizophrenia are often long and linear ([Exhibit 6](#)). Of note, slope-change in Weekly TRx is often associated with indication expansion (i.e. NDA in schizophrenia, sNDA in bipolar depression), as demonstrated with Latuda and Caplyta's sNDA in bipolar depression. Of note, Vraylar launched with an approval in both Schizophrenia and Bipolar disorder. Cobenfy ([Exhibit 7](#)) is currently tracking behind Latuda and Vraylar for its full year of launch, and remains approved in only one indication, schizophrenia.

Exhibit 6: Schizophrenia launches are typically long and linear (barring indication expansion)

Source: Morgan Stanley Research, IQVIA

Exhibit 7: Year one TRx for recent major antipsychotic launches

Source: Morgan Stanley Research, IQVIA

4. How do muscarinics differ from typical/atypical antipsychotics? Current

antipsychotics vary by the extent to which they inhibit dopamine vs. serotonin receptors in the brain (low K_i values for D2 vs. 5-HT_{2A}, see [Exhibit 8](#)). In contrast, muscarinic agonists (e.g. Cobenfy and MapLight's ML007-C-MA) act on acetylcholine receptors (M1/M4), not dopamine or serotonin. This means muscarinics avoid class effects which limit compliance (see below). 50-55% are not adherent to their currently prescribed antipsychotic.

Exhibit 8: Typical antipsychotics inhibit D2 more than 5-HT_{2A} (low K_i values indicate high binding and therefore strong inhibition), opposite to atypical antipsychotics

Receptor Occupancy, K_i (nM) <i>low K_i values = strong binding</i>	Dopamine (D1)	Dopamine (D2)	Serotonin (5-HT _{2A})	Serotonin (5-HT _{2C})	Serotonin (5-HT ₆)	Serotonin (5-HT ₇)	Acetylcholine (MACHR)
Typical antipsychotics							
Haldol (Haloperidol)	270	1.4	< 25.0	UL	UL	380	4,669
Thorazine (Chlorpromazine)	71	2.6	< 12.0	6	4	21	67
Atypical antipsychotics							
Abilify (Aripiprazole)	1,960	0.7	< 8.7	76	570	10	6,780
Seroquel (Quetiapine)	4,240	310.0	> 120.0	3,820	UL	290	1,020
Zyprexa (Olanzapine)	250	17.0	> 1.9	7	3	120	26
Invega (Paliperidone)	670	4.0	> 0.3	71	UL	1	3,570
Risperdal (Risperidone)	620	3.3	> 0.2	63	420	2	UL
Geodon (Ziprasidone)	330	9.7	> 0.3	13	76	5	UL
Clozaril (Clozapine)	540	150.0	> 3.3	13	4	21	34

UL = upper limit, no binding activity (K_i values >10,000)

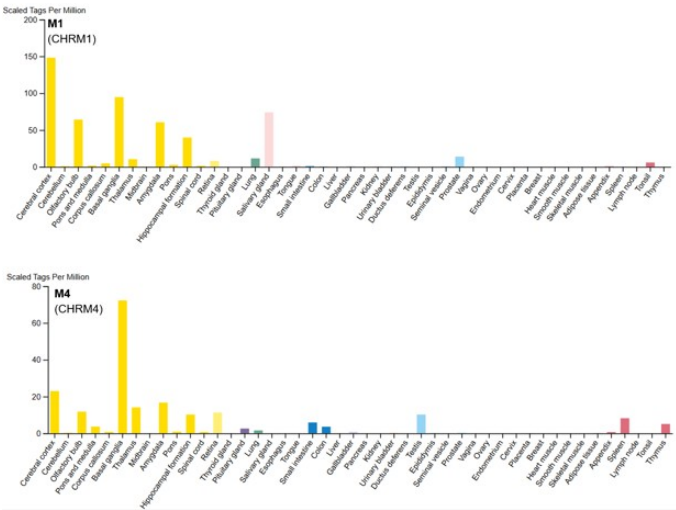
Source: Morgan Stanley Research, Kusumi et al

- Typical antipsychotics strongly inhibit dopamine (low K_i values for D2, see [Exhibit 8](#)) and as a result, dopamine-induced movement disorders like [extrapyramidal symptoms](#) are common with typical antipsychotics. They also cause tardive dyskinesia, with [reports](#) suggesting an annualized incidence of 6% for typical antipsychotics in schizophrenia (vs 4% for atypical antipsychotics).
- Atypical antipsychotics strongly inhibit serotonin (low K_i values for 5-HT_{2A}), leading to [serotonin-induced weight gain](#) (common with atypicals). As an aside, Abilify is the only approved atypical with stronger inhibition of dopamine (which originally positioned Abilify to be perceived as "weight-neutral"). However, we note that Abilify shows low K_i values on both D2 and 5-HT_{2A}, and reports suggest clinically meaningful weight gain (8-11kg) in the first few months, in addition to weight loss following discontinuation.

5. What are the drawbacks to muscarinics? Muscarinics have an inherent trade-off. They lower psychosis by activating M1 and M4 receptor in the brain (yellow bars, see [Exhibit 9](#)), but do so at the expense of binding M2 and M3 in the body (non-yellow bars, see [Exhibit 10](#)). As evidence, [Cobenfy](#) primarily targets M1 and M4, but also shows activity on M2, M3, and M5. So, muscarinics outside of the brain cause **procholinergic** adverse events. To counteract this, pairing with a peripheral anticholinergic (or PAC) like trospium blocks procholinergic AEs, but too much blockade leads to **anticholinergic** AEs ([Exhibit 11](#)). As a result, muscarinics must balance a narrow therapeutic index for maximizing reduction of psychosis while ensuring lower rates of cholinergic side effects. As discussed next, we believe this setup ideally positions MLO07-C-MA to have competitive and potentially best-in-class profile on both efficacy and safety.

- Single receptor subtype-selective agonists (e.g., M4-selective direclidine/NBI-568) are an alternate strategy to a broad muscarinics, but there are five receptor subtypes, and their structures are near identical (active site homology is conserved across M1-M5). This creates technical hurdles for developing single receptor subtype-selective agonists (e.g. M4-selective agonist [emraclidine](#) showed >800x preference for M4 over other M subtypes).

Exhibit 9: M1 and M4 are highly expressed in the brain (yellow bars), driving the reduction in psychosis



Source: Morgan Stanley Research, Human Protein Atlas

Exhibit 11: Summary of procholinergic and anticholinergic side effects

Procholinergic Side Effects

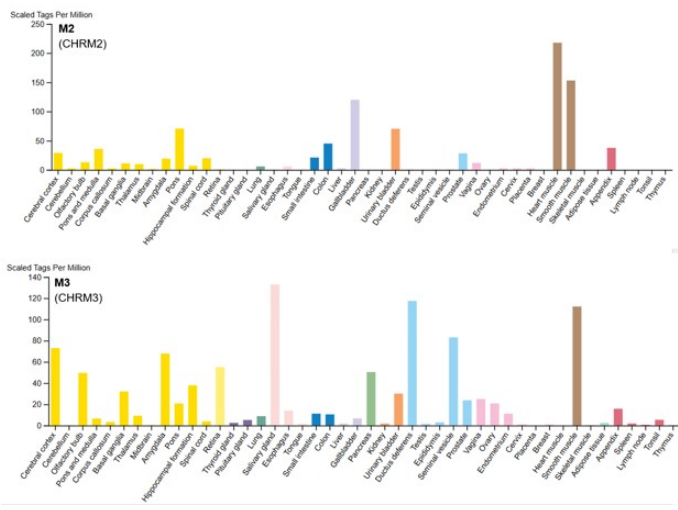
Mechanism: Activation of muscarinic receptors outside of the brain.

Typical symptoms:

- Excess salivation
- Nausea and vomiting
- Urinary urgency
- Diarrhea
- Bradycardia (slow heart rate)
- Excess sweating

Source: Morgan Stanley Research

Exhibit 10: M2 and M3 are broadly expressed outside the brain (non-yellow bars), leading to adverse events



Source: Morgan Stanley Research, Human Protein Atlas

Anticholinergic Side Effects.

Mechanism: Inhibition of muscarinic receptors outside of the brain.

Typical symptoms:

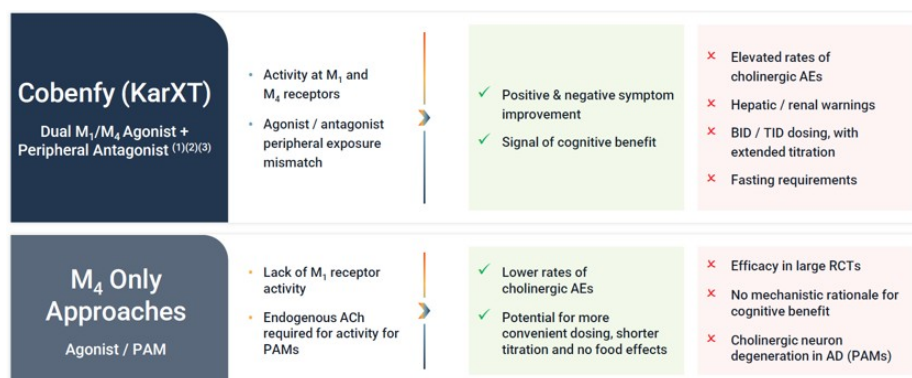
- Dry mouth
- Constipation
- Urinary retention
- Blurred vision
- Tachycardia (fast heart rate)
- Decreased sweating

Why is ML007-C-MA competitive, despite the lag?

ML007-C-MA is an optimized broad muscarinic + PAC. There are two strategies ([Exhibit 12](#)) for targeting muscarinic receptors; **(1) Broad muscarinic agonists** (with preference for M1/M4) + a PAC (e.g., Cobenfy, ML007-C-MA) vs. **(2) Selective modulators** (e.g., emraclidine, direclidine). Despite ML007-C-MA's delay to commercialization relative to peers (outlined below), we see several reasons that favor the broad muscarinic approach.

- MapLight's (MPLT) **ML007-C-MA** Phase 2 ZEPHYR topline results are expected in schizophrenia in 2H26, followed by Phase 2 VISTA topline results in Alzheimer's disease psychosis in 2H27.
- Bristol Myers' (BMY) **Cobenfy** is approved in schizophrenia (Phase 3 ADEPT studies in ADP are expected to read out in 4Q25 and next year).
- Neurocrine's (NBIX) **Direclidine (NBI-568)** is in ongoing Phase 3 studies for schizophrenia (expected to readout in 2027). Additional dual (yet apparently selective for M1/M4 and not M2, M3, or M5) programs or preferring programs are in Phase 1 development.

Exhibit 12: Muscarinic target approaches rely on a broad muscarinic + PAC, or a selective approach



Source: Company reports

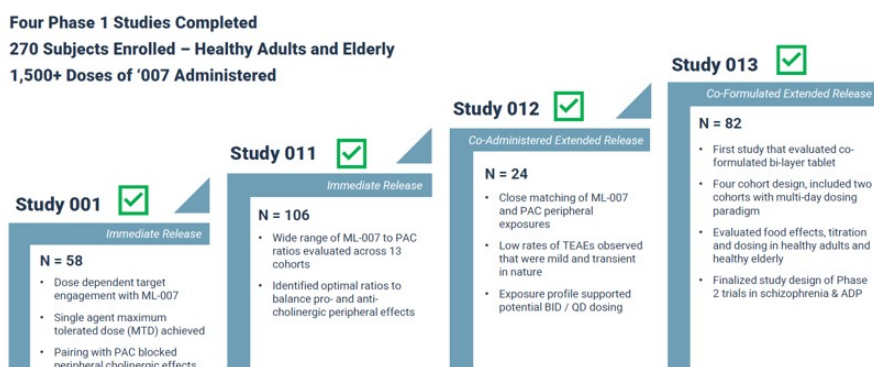
1. **Broad muscarinics** like [Xanomeline \(without trospium\)](#) reduce psychosis (by binding M1 and M4 in the brain), but causes pro-cholinergic adverse events (from binding of xanomeline to M2, M3 in periphery). The inclusion of trospium helps block procholinergic AEs, but too much trospium can lead to anticholinergic AEs.
 - a. Regulators treat fixed-combination drug products (like ML-007-C-MA) as dual molecular entities and therefore require sponsors to substantiate claims on each product's efficacy, safety and tolerability, including the amount and frequency of each component in combination.
2. **Selective modulators** like emraclidine and direclidine are selective for M4 only, and therefore do not require a peripheral PAC, creating in our view, a "double-edged sword". Their selectivity allows for use with no PAC, avoiding potential for anticholinergic events. That said, this selectivity likely results in less broad and potent muscarinic agonist, limiting potential for maximal efficacy/agonism of M receptor types, and downstream PANSS score reduction.
 - a. Emraclidine failed to demonstrate compelling clinical efficacy in its registrational EMPOWER studies. That said, NBI-568 (now direclidine) yielded

Phase 2 PANSS scores that were somewhat below that of Cobenfy (but still highly competitive), and the program is in ongoing Phase 3 evaluation.

ML007 has optimized PK/PD

ML007 Pharmacokinetics (PK). ML007C-MA is a fixed-dose of ML007 (M1/M4 agonist) + PAC fesoterodine. MapLight optimized ML007-C-MA across 270 healthy adults and elderly subjects and 4 Phase 1 studies ([Exhibit 13](#)). We provide highlights from each study below. In summary, the results demonstrate highly synchronous ML007:PAC exposures stayed within target plasma ratios of ML-007:PAC of 100:1 to 600:1 ([Exhibit 14](#)). With no deviation outside target range, this stable, synchronous ratio of muscarinic:anticholinergic demonstrated a well-tolerated safety profile, enabling higher CSF exposures ([Exhibit 15](#)). These studies have yielded optimized go-forward Phase 2 doses for ML007-C-MA in both schizophrenia ([Exhibit 16](#)) and Alzheimer's disease psychosis ([Exhibit 17](#)).

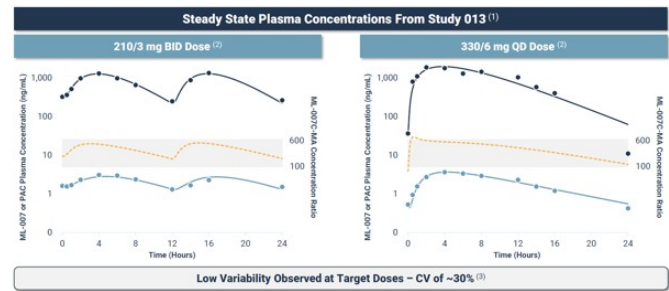
Exhibit 13: ML007-C-MA Phase 1 development for optimized ratios and formulation



Source: Company reports

- **Study 001.** 2 cohorts, evaluated ML-007 oral solution (0.8 mg, 2.5 mg, 8.2 mg, 16 mg, 32 mg and 49 mg, with 32mg identified as maximum tolerated dose), and 6 mg PAC extended-release (ER).
- **Study 011.** 13 cohorts, evaluated 22 different dosing paradigms, with a target plasma ratio of ML-007:PAC of 100:1 to 600:1 established ([Exhibit 14](#)). Anticholinergic events (dry mouth, tachycardia, feeling hot, and dry eye) occurred at ratios <100:1, and procholinergic events (nausea, vomiting, diarrhea, hyperhidrosis, and hypersalivation) occurred at ratios >600:1.
- **Study 012.** Evaluated single escalating dose of ML007 ER vs. ML007 ER + PAC ER.
- **Study 013.** Evaluated pharmacokinetics, safety and tolerability of ML-007C-MA in healthy adults and elderly participants. Participants were dosed for up to 14 days (for elderly volunteers, a 2-7 day titration period was observed followed by dosing at target doses for 7 days), with the 100:1 to 600:1 plasma ratio maintained throughout ([Exhibit 14](#)). CSF exposures were above anticipated clinically relevant levels ([Exhibit 15](#)) with both once- and twice-daily dosing regimens. Based on results in healthy adults, the 210mg/3mg BID dose and 330/6mg QD dose were selected as the go-forward dose for Phase 2 ZEPHYR studies for schizophrenia ([Exhibit 16](#)), and based on results in healthy elderly, 210/3 mg BID with 2-7day titration was selected for Phase 2 VISTA ([Exhibit 17](#)).

Exhibit 14: ML007 Phase 1 Study 013 plasma results demonstrate no excursions from the target 1:600 ratio



Source: Company reports

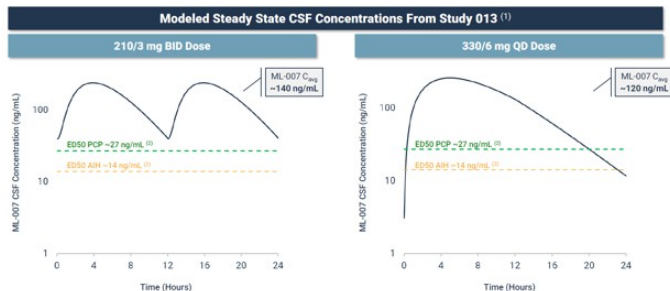
Exhibit 16: Phase 1 Study 013 results in non-elderly support dosing for Phase 2 ZEPHYR

	Placebo Combined (N=8)	165/3 mg BID (N=6)	210/3 mg BID (N=6)	270/6 mg QD (N=6)	330/6 mg QD (N=6)
Subjects With Any TEAE ⁽¹⁾	2 (25%)	6 (100%)	3 (50%)	3 (50%)	4 (67%)
Cholinergic Classification – Subjects (%)					
Procholinergic	0 (0%)	4 (67%)	3 (50%)	2 (33%)	4 (67%)
Anticholinergic	1 (13%)	3 (50%)	0 (0%)	1 (17%)	0 (0%)
Other	1 (13%)	5 (83%)	3 (50%)	3 (50%)	3 (50%)
% of TEAEs by Severity					
Mild	100%	100%	95%	93%	86%
Moderate	0%	0%	5%	7%	14%
Severe	0%	0%	0%	0%	0%

Phase 2 Target Doses for Schizophrenia

Source: Company reports

Exhibit 15: ML007 Phase 1 Study 013 CSF results demonstrate no excursions from the target 1:600 ratio



Source: Company reports

Exhibit 17: Phase 1 Study 013 results in elderly support dosing for Phase 2 VISTA

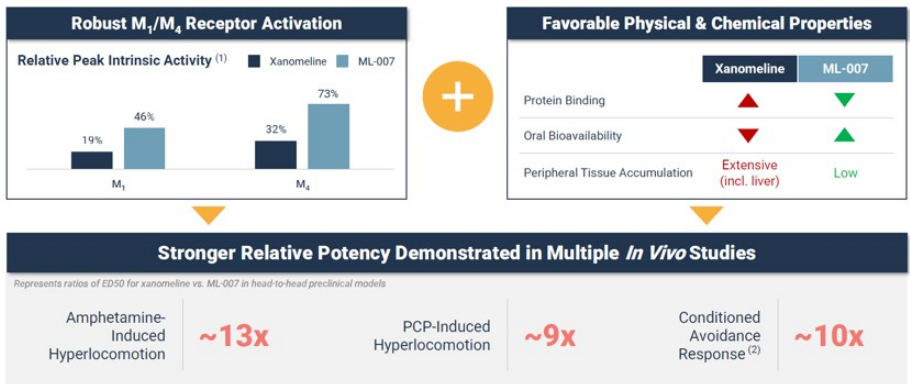
	Placebo Combined (N=7) ⁽²⁾	165/3 mg BID (with 2d titration) (N=6)	210/3 mg BID (with 2-7d titration) (N=11) ⁽³⁾	330/6 mg QD (with 7d titration) (N=6) ⁽⁴⁾
Subjects With Any TEAE ⁽¹⁾	5 (71%)	4 (67%)	8 (73%)	6 (100%)
Cholinergic Classification – Subjects (%)				
Procholinergic	1 (14%)	2 (33%)	6 (55%)	6 (100%)
Anticholinergic	1 (14%)	1 (17%)	4 (36%)	0
Other	3 (43%)	3 (50%)	8 (73%)	6 (100%)
% of TEAEs by Severity				
Mild	100%	100%	93%	72%
Moderate ⁽²⁾	0%	0%	7%	28%
Severe	0%	0%	0%	0%

Phase 2 Target Dose for ADP

Source: Company reports

ML007 Pharmacodynamics (PD). For ML007, we believe its targeting strategy + PAC is likely to deliver more potent agonism and therefore greater reductions in psychosis (as measured via PANSS). While clinical data to compare **ML007** vs. **(2) Selective ortho/allosteric modulators** are not readily available, within the **(1) Broad muscarinic** approach, preclinical data suggest ML007 has more potent M1 and M4 activation vs Cobenfy (Exhibit 11). In head-to-head preclinical models (amphetamine and PCP-induced movement, avoidance response), a 10-13x lower ML007 drug exposure was needed for efficacy vs. Cobenfy.

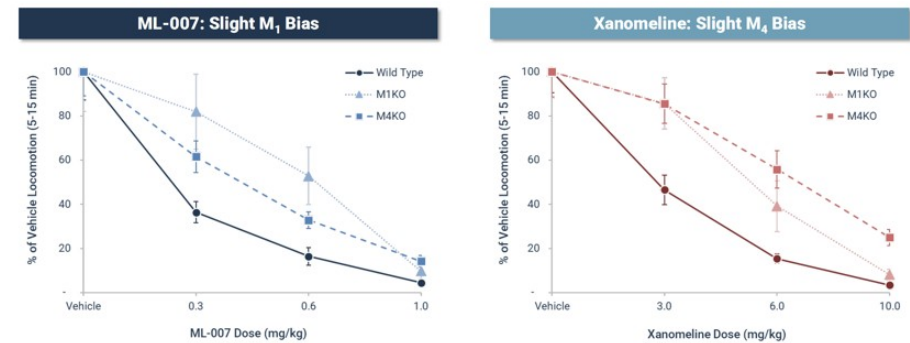
Exhibit 18: ML007 preclinical assays showcase a highly potent M1/M4 agonist



Source: Company reports

ML007 M1/M4 preference vs. Xanomeline. Both ML007 and Xanomeline show strong binding activity on M1 and M4 receptors. [Preclinical results](#) (in vitro assays of human M1 and M4 receptor in mouse cells) demonstrate ML007 is M1 preferring, whereas Xanomeline shows a slight preference for M4>M1 ([Exhibit 19](#)).

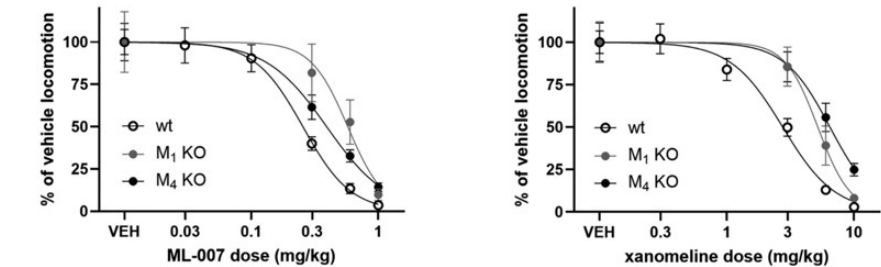
Exhibit 19: ML007 has stronger preference for M1 while Xanomeline prefers M4



Source: Company reports

ML007 potency and bioavailability in brain. Of note, ED50 values for ML007 are meaningfully lower vs. Xanomeline in this preclinical head-to-head ([Exhibit 20](#)), supportive of ML007's increased potency vs Xanomeline. These [preclinical results](#) also note that on bioavailability, "ML-007 CSF concentration reached 115 ng/ml (or 698 nM) following 0.3 mg/kg IP injection, while the xanomeline CSF concentration was below the level of quantification (BLOQ = 0.5 ng/ml or <1.8 nM xanomeline) following 3 mg/kg IP injection. Thus, the molar CSF Cmax concentration of ML-007 was more than 380 times that of the molar CSF Cmax of xanomeline". These data are in addition to the potency described above in in vitro assays, and in totality, they demonstrate that ML007 is both more CSF penetrant and more potent. This should allow for lower ML007 doses needed to achieve higher brain exposures vs Xanomeline, and lead to less ML007 outside of the brain. Indeed, ML007's design increases CNS efficacy while lowering chances for peripheral accumulation and cholinergic AEs.

Exhibit 20: Preclinical head-to-head ED50 values demonstrate ML007 is more potent and M1 preferring



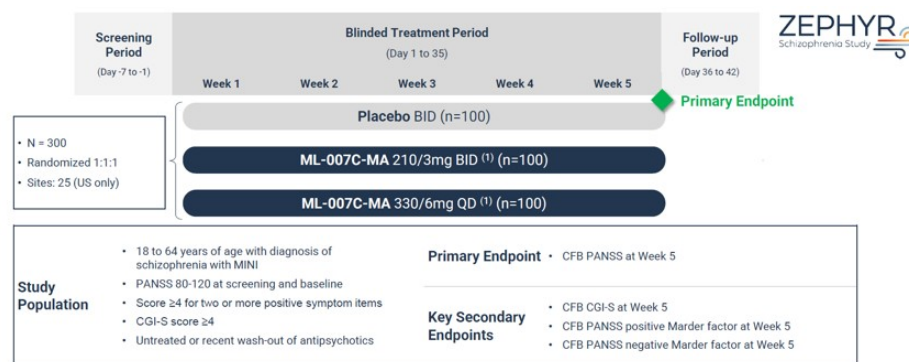
	Mouse strain	ED ₅₀ (mg/kg)	95% CI (mg/kg)
ML-007	WT	0.26	0.24 – 0.27
	M ₁ KO*	0.58	0.37 – 0.81
	M ₄ KO*	0.39	0.36 – 0.43
Xanomeline	WT	2.75	2.20 – 3.36
	M ₁ KO*	5.20	4.82 – 5.58
	M ₄ KO*†	6.47	6.04 – 6.93

Source: Chatterjee et al 2025

ML007 Phase 2 ZEPHYR study design

ML007-C-MA is evaluating a once and twice daily dose in schizophrenia with topline results in 2H26. Phase 2 ZEPHYR is expected to enroll ~300 patients evaluating placebo BID vs. ML007-C-MA 210/3mg BID or 330/6mg QD ([Exhibit 21](#)). The primary endpoint is change from baseline in Positive and Negative Syndrome Scale (PANSS) total score at Week 5. The study is enrolling patients with a baseline score of 80-120. For context, a PANSS score of 58 correlates to mild severity, increasing to 95 for markedly ill and 116 for severely ill schizophrenia patients ([Leucht et al](#)). ZEPHYR enrolls from a limited number of clinical trial sites (<25) and in US only for mitigation of placebo response. The established precedent suggests quality and number of trial sites (and ex-US enrollment) are meaningful upward drivers of placebo response. For reference, KarXT Phase 2 EMERGENT-1 enrolled from 12 sites, direclidine enrolled from 15 sites, and emraclidine Phase 1b was run across only 5 trial sites.

Exhibit 21: ML007-C-MA Phase 2 ZEPHYR study for schizophrenia



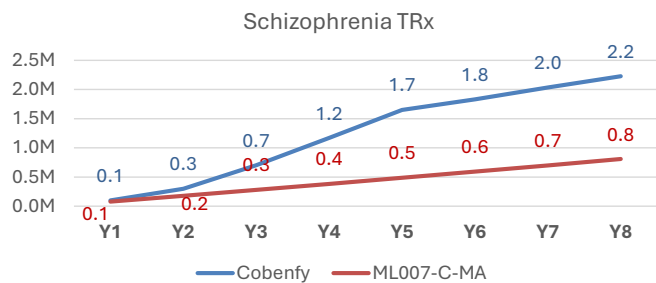
Source: Morgan Stanley Research

Model Assumptions

ML007-C-MA scripts uptake (relative to Cobenfy). For context, we note that in the original 1997 LLY Phase 2 studies in Alzheimer's disease (conducted with xanomeline only, no tropism), poor tolerability led to a dose-dependent increase in discontinuation in placebo (35%) vs. xanomeline 75mg (19%), 150mg (48%), and 225mg (59%). Despite being in a different patient population, the dose-dependence on discontinuation rates aligns with AE profile and suggests that tolerability is a major driver of whether patients stay on therapy. With ML007-C-MA precision matched PK, we expect lower rates of both pro- and anticholinergic adverse events. Importantly, we think this should translate to higher rates of patient compliance.

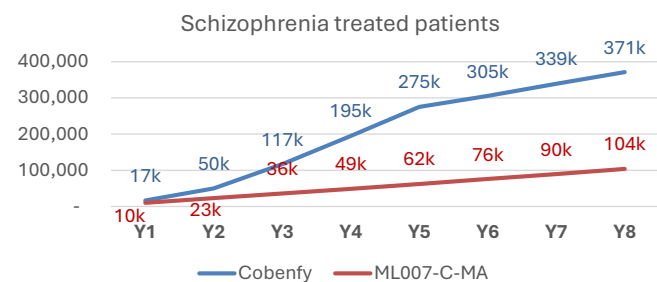
- For ML007, we model 8 TRx per year and ~104K patients ([Exhibit 22](#)) on drug 8 years into the launch, net of 832,000 or 0.8M ([Exhibit 23](#)) in peak ML007-C-MA TRx for schizophrenia.
- For Cobenfy, MSe imply 6 TRx per year and ~370k patients on drug 8 years into the launch, net of 2,226,000 or 2.2M in peak Cobenfy TRx for schizophrenia (MSe estimates for BMV's Cobenfy from Terence Flynn).

Exhibit 22: ML007-C-MA schizophrenia TRx vs. Cobenfy



Source: Morgan Stanley Research

Exhibit 23: ML007-C-MA schizophrenia patients treated vs. Cobenfy



Source: Morgan Stanley Research

We assume 30% of currently treated patients as the addressable population. We estimate 2.8M schizophrenia patients at 2029 launch year, 1.7M seeking treatment, and 30% fully refractory to current regimens, representing a 0.5M addressable population (Exhibit 24). We note that the additional 30-45% are not adequately controlled, and they remain upside to our ML007-C-MA estimates (pending commercial performance from Cobenfy). On price, our \$24k ML007 launch price assumes a modest premium over Cobenfy (~ \$22K currently), and net of ~8 TRx per patient per year (12 months * 65% compliance). We model ~20% peak market share and risk adjusted revenues of \$850M (\$1.4B unadjusted).

Exhibit 24: Market model for ML-007C-MA for treatment of psychosis in schizophrenia

Schizophrenia - US		2025A	2026A	2027A	2028A	2029A	2030A	2031A	2032A	2033A	2034A	2035A	2036A
US adult population (M)	0.9%	269.2	271.5	273.8	276.2	278.5	280.9	283.3	285.7	288.1	290.5	293.0	295.5
Schizophrenia prevalence (M)	1%	2.7	2.7	2.7	2.8	2.8	2.8	2.8	2.9	2.9	2.9	2.9	3.0
Treated schizophrenia patients (M)	60%	1.6	1.6	1.6	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.8	1.8
Patients refractory to atypicals	30%	484,619	488,738	492,893	497,082	501,307	505,568	509,866	514,200	518,570	522,978	527,424	531,907
Market share						2.0%	4.5%	7.0%	9.5%	12.0%	14.5%	17.0%	19.5%
Patients treated						10,026	22,751	35,691	48,849	62,228	75,832	89,662	103,722
Annual gross price (\$)						24,000	24,720	25,462	26,225	27,012	27,823	28,657	29,517
%Y/Y							3%	3%	3%	3%	3%	3%	3%
Annual net price (\$)						18,000	18,416	18,842	19,276	19,719	20,171	20,633	21,105
Gross to net						25%	26%	26%	27%	27%	28%	28%	29%
Compliance	65%												
ML-007c unadjusted US sales (\$M)						\$117	\$272	\$437	\$612	\$798	\$994	\$1,203	\$1,423
ML-007c risk-adj. US sales (\$M)	60%					70	163	262	367	479	597	722	854

Source: Morgan Stanley Research

Debate 2: Muscarinics in Alzheimer's: Is there a coming paradigm shift?

Overview: Alzheimer's disease psychosis (ADP) has been among the most arduous indication for CNS drug development (we have FDA approved disease-modifying therapies before an FDA approved antipsychotic). As compared to schizophrenia, patients with Alzheimer's disease (AD) are older, frailer, and suffer from neurodegeneration. This makes efficacy hard to achieve (brain cells, which are the target of antipsychotics, are readily dying), and makes safety even more paramount for patient adherence. In this market, above all, there is room for any effective therapy and therefore several winners.

Street's take: Muscarinics are poised to expand their addressable market, with several shots on goal. BMY ADEPT studies in ADP (ADEPT-1, 2, 4, and 5) are expected to readout, starting in 4Q25 with ADEPT-2. MapLight's MLO07-C-MA Phase 2 ZEPHYR results in schizophrenia will readout in '2H26, but Phase 2 VISTA topline results in Alzheimer's disease psychosis are expected in '2H27.

Our take: Alzheimer's disease psychosis (ADP), if treated, would be an immediate blockbuster. Despite the clinical risk, there are several reasons to be bullish on muscarinics for ADP. 40% of the ~8M US patients suffering from AD currently have psychotic symptoms (3.2M patients), but reports suggest 97% will have a psychotic episode in the course of their disease. As evidence, off-label antipsychotic use is rampant- estimated in 30% of all AD patients (3.2M patients). Prior Eli Lilly Phase 2 data show proof of clinical efficacy with muscarinics in ADP, but were limited by tolerability and dose-dependent discontinuations.

Our view on Alzheimer's disease psychosis (ADP) and what's needed for success. Put plainly, we believe that all that's missing for success in ADP with muscarinics is a well-tolerated profile to ensure compliance. While there are no FDA-approved treatments for ADP, many patients are using traditional antipsychotics off-label, and there are already data to support clinical benefit with muscarinic agonists in lowering psychosis in ADP. Xanomeline (without trospium) was previously evaluated in AD in a Phase 2 study run by Eli Lilly ([Bodick et al 1997](#)). The results clearly demonstrate that xanomeline delivers clinically meaningful reduction in hallucinations and delusions (as early as Week 3), however, they also clearly demonstrate a dose-dependence on procholinergic events leading to discontinuation. Indeed, the prior Eli Lilly study demonstrates that compelling clinical efficacy was already demonstrated with a muscarinic in ADP, but the program was limited by compliance at Week 4.

Reasons we are bullish on the potential for muscarinics. In our view (and given the above), positive clinical data in ADP should therefore be largely expected. Indeed, the 1997 Phase 2 results were positive (up until discontinuation). In other words, had patients been able to tolerate to xanomeline alone (the study was designed for 6-months of duration vs. the more usual 5-6 week in neuropsychiatry), this could have supported potential for a

positive readout (and subsequent filing). As a result, we think the precedent positions MapLight's ML007-C-MA as meaningfully differentiated in a key area (reduction of pro and anticholinergic events). Ultimately, this should increase compliance (relative to the 1997 studies of xanomeline alone, but also relative to Cobenfy's potential profile in ADEPT), and therefore further increase chances for positive topline results from ML-007-MA Phase 2 VISTA topline in 2H27.

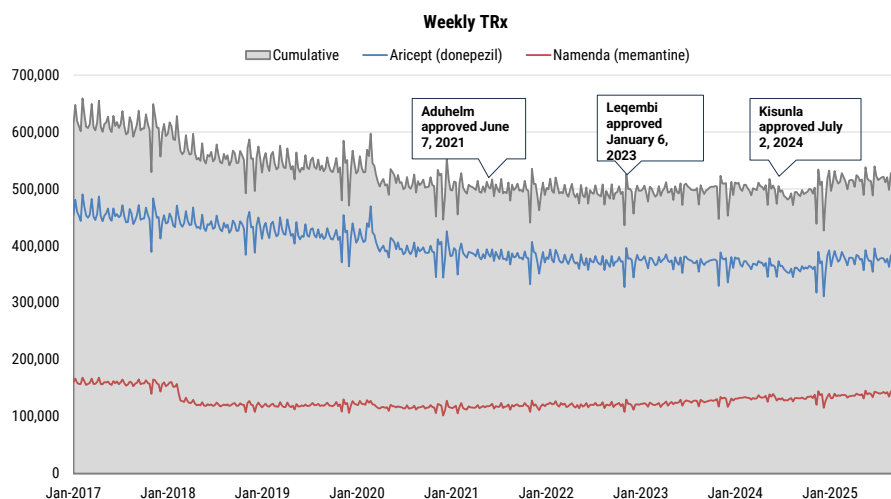
1. Expansion of muscarinic agonists in ADP would signal a paradigm shift in the treatment of psychosis. There is a growing need for the 6-8M US adults suffering from Alzheimer's disease (AD), 40% of whom have psychotic events and a third of whom are currently on off-label antipsychotics. This implies in excess of 3M addressable ADP patients. We note up to 97% of patients will experience a bout of psychosis throughout the course of their disease.

2. Disease-modifying therapies have considerable room for efficacy. Current industry focus is on optimizing amyloid clearance, whereas patients and their providers are often focused on memory loss and cognitive deficits. Amyloid plaques are the hallmark pathology of AD, and patients are treated with A β antibodies (Leqembi, Kisunla) to directly lower amyloid, however they are limited in their ability to slow cognitive decline.

3. As evidence, prescribing patterns for AD have remained fairly steady despite approval of multiple treatments. Despite several years into the launch for A β therapies, the AD market remains dominated by Aricept (donepezil) and Namenda (memantine), indicated for treating dementia (Exhibit 25). Namzaric is a combination of the two. These therapies remain standard-of-care for treating cognitive deficits in AD, but do little to address the psychosis, which is highly prevalent.

- Of note, aricept is an acetylcholinesterase inhibitor (AChEI) that prevents the breakdown of acetylcholine, increasing its tone in the brain (and therefore suggests some commonalities between aricept and muscarinic agonism). Other common AChEI include rivastigmine and galantamine.

Exhibit 25: Weekly scripts for mainstay therapies for treatment of cognitive aspects of AD

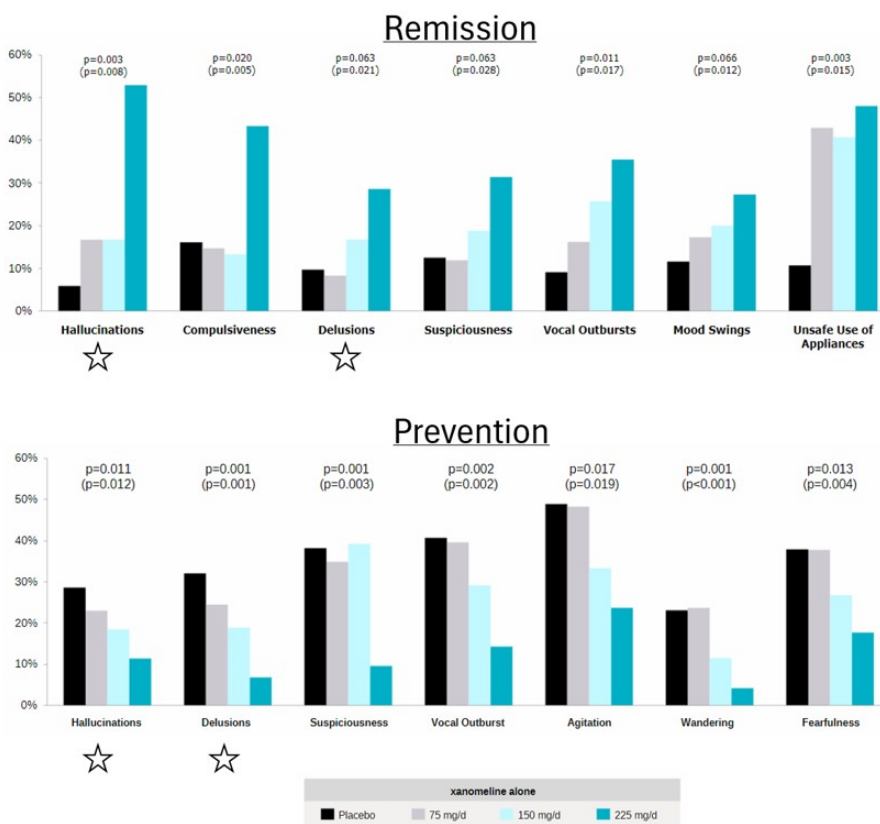


Source: Morgan Stanley Research, IQVIA

4. There are already data to support clinical benefit in ADP. Recall that Xanomeline (without trospium) was previously evaluated for the treatment of psychosis in AD in a Phase 2 study, where results clearly demonstrate that xanomeline delivers clinically meaningful reduction in psychosis. However, they also show dose-dependence on discontinuation. Indeed, the 1997 Lilly study was conducted with xanomeline (without trospium), and were the first clinical demonstration of procholinergic adverse events (occurring in up to 76% of the Lilly study). In our view, this suggests muscarinic development in ADP is largely de-risked (relative to the early Lilly studies), given the addition of PAC. Indeed, assuming the PAC can ensure tolerability, we believe the efficacy component is already de-risked.

- **Efficacy.** Recall that psychosis is defined as recurring hallucinations and delusions. On **remission**, only 7% (4/59) of the 225mg treatment arm developed psychosis vs. 32% (16/56) of patients on placebo who developed psychosis ([Exhibit 26](#)). On **prevention**, by Week 3, 53% of patients on 225mg xanomeline showed a clinical prevention of psychosis, vs. 6% (1/17) with placebo ([Exhibit 26](#)).

Exhibit 26: Xanomeline already demonstrated its efficacy in ADP for both remission and prevention of psychosis (see hallucinations and delusions)

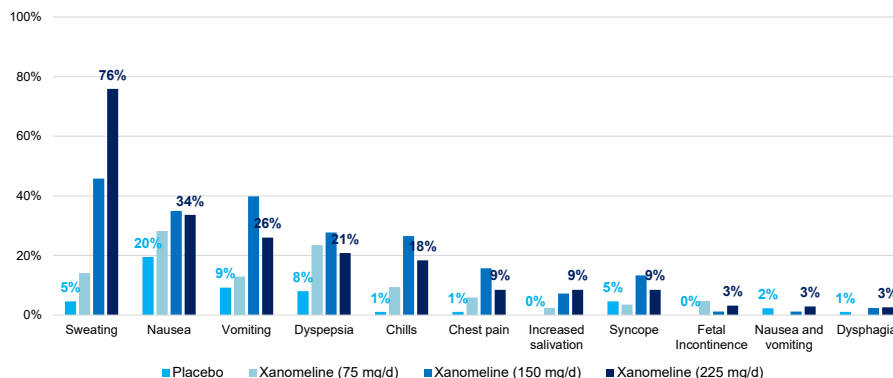


Source: Morgan Stanley Research, [Bodick et al 1997](#)

- **Safety.** Procholinergic events (sweating, nausea, and vomiting, etc.) were observed and clearly demonstrate dose-dependence ([Exhibit 27](#)). On **discontinuation**, by Week 4, 36% (28/78) of patients receiving 225mg xanomeline had already discontinued treatment due to side effects, namely procholinergic events. By end of study, poor tolerability led to a dose dependent increase in discontinuation in

placebo (35%) vs. xanomeline 75mg (19%), 150mg (48%), and 225mg (59%).

Exhibit 27: Xanomeline showed dose-dependent increases in procholinergic adverse events, notably sweating, nausea, and vomiting



Source: Morgan Stanley Research, Bodick et al 1997

- As an aside, we believe these data are likely to influence expectations heading into BMY ADEPT studies. We see the **Prevention data** (Exhibit 26, bottom panel) are more in line with **ADEPT-1 and -4**, whereas we think the **Remission data** (Exhibit 26, upper panel) are likely to influence expectations heading into **ADEPT-2 and -5** (Exhibit 28). We also note that BMY has stated that two of the four ADEPT studies (ADEPT-1, 2, 4, or 5) must show positive results for them to pursue regulatory approval for Cobenfy in Alzheimer's disease psychosis.

Exhibit 28: Cobenfy Phase 3 ADEPT trial design summary

Trial	Primary Goal	Endpoint Type	Topline Results Expected
ADEPT-2	Acute symptom control	Change from Baseline in NPI-C: H+D	4Q25
ADEPT-1	Relapse prevention	Time from randomization to relapse	2026
ADEPT-4	Relapse prevention	Time from randomization to relapse	2026
ADEPT-5	Acute symptom control	Change from Baseline in NPI-C: H+D	TBD

Source: Morgan Stanley Research, Company reports

5. Patients have 3-8 years from diagnosis, and are more often frail, creating a high bar on safety. There is usually considerable time from first symptom presentation to diagnosis. Patients only are diagnosed once clinical aspects are severe enough to impact daily life. As a result, clinical intervention is usually at later, more advanced disease stages, resulting in a more frail population. Indeed, AD remains a leading cause of death in the United States, ranking as the sixth leading cause for people 65 and older and the seventh for all U.S. adults. Disease pathology disproportionately affects patients diagnosed in later years

- For those diagnosed at age 65, studies note a median survival of 8.3 years (implying a 73.3 year lifespan) vs. those diagnosed at age 90, with a median survival of 3.4 years (implying a 93.4 year lifespan).

6. Therefore, compliance matters most for evaluating MapLight's ML007-C_MA potential in ADP. Investors have a tendency to focus on efficacy as the primary factor dictating clinical outcome and therefore regulatory approval. However, the clinical precedent would suggest that xanomeline should have had strong chances for regulatory

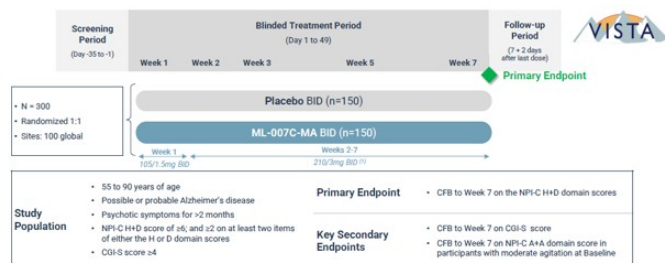
approval, should patients had been able to complete the 6-month study.

7. MLO07-C-MA's study and drug design should ensure better tolerability vs Cobenfy.

While the original 1997 studies were limited by procholinergic events, we expect that the co-formulation of xanomeline + trospium should help to address this, but with room for improvement. We see several avenues for differentiation vs Cobenfy (see below), which we believe can help mitigate readthrough to MLO07-C-MA Phase 2 VISTA (should Phase 3 ADEPT studies read out negatively).

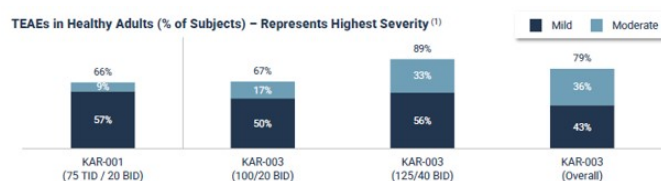
- **Study design.** MLO07-C-MA Phase 2 VISTA ([Exhibit 29](#)) is designed to evaluate efficacy versus placebo in reducing hallucinations and delusions in patients with ADP. As compared to the ADEPT studies, we note MLO07-C-MA Phase 2 VISTA employs a shorter duration (7 weeks), which could help for lowering attrition and dropout risk in frail AD patients.
- **Safety.** We think there is strong scientific merit supporting potential for lower rates of AEs with MLO07-C-MA's (as a result of its optimized target plasma ratio of ML-007:PAC of 100:1 to 600:1, see [Exhibit 14](#), [Exhibit 15](#)). Indeed, MLO07-C-MA has demonstrated to maintain a stable ratio, which we think can lead to a meaningfully differentiated PK/PD profile (and [Exhibit 20](#)). For context, both pro and anticholinergic events were observed in KartXT Phase 1. >40% on elderly subjects on KarXT in Phase 1 showing moderate adverse events ([Exhibit 30](#)), plus an additional >30% with mild events. We believe safety events are likely the result of a deviation outside the target plasma range for Cobenfy.

Exhibit 29: MLO07-C-MA Phase 2 VISTA study design



Source: Company reports

Exhibit 30: KarXT in Phase 1 demonstrated an enduring adverse even profile, the majority of which were mild-to-moderate



Source: Company reports

- **Tolerability and drug exposure.** Unlike Cobenfy, MLO07-C-MA does not require a fasted state. Considering the above, we therefore expect improved tolerability with MLO07-C-MA to translate to increased drug exposure while maintaining compliance.
- **Efficacy.** Recall that MLO07-C-MA is more potent M1/M4 agonist, and given the potential for increased compliance outlined above, we believe this can ensure better efficacy (relative to BMY ongoing ADEPT trials). In our view, this should help mitigate downside risk, and introduces two scenarios for the Phase 2 VISTA topline outlined below ([Exhibit 31](#)).

Exhibit 31: Readthrough for ML007-C-MA Phase 2 VISTA in ADP pending outcomes from Cobenfy ADEPT studies

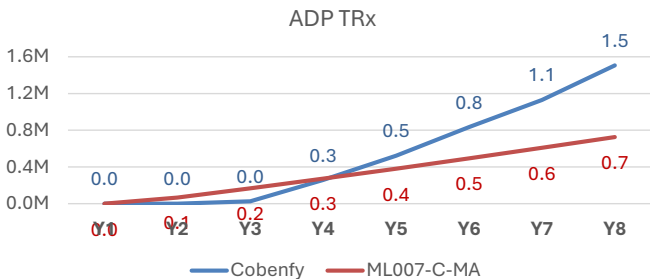
Scenario	Outcome	Readthrough for ML007-C-MA Phase 2 VISTA in ADP
Scenario A	Positive ADEPT Topline	De-risked expansion potential for muscarinics in ADP, clinical benchmark and clear regulatory precedent can be established.
Scenario B	Negative ADEPT Topline	Market discount on probability-of-success for ADP expansion with muscarinics. But, ML007-C-MA's differentiation (improved potency, tolerability) could allow for increased compliance and patient drug exposure.

Source: Morgan Stanley Research

Model Assumptions

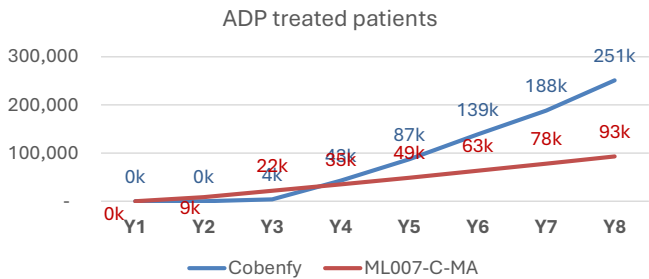
ML007-C-MA uptake in ADP (relative to Cobenfy). For ML007, we model 8 TRx per year (Exhibit 31) and 93K patients at peak on ML007-C-MA at peak (Exhibit 32). For context, BMY MSe implies 1.5M ADP TRx, derived from 250k patients. (MSe estimates for BMY's Cobenfy from Terence Flynn).

Exhibit 32: ML007-C-MA ADP TRx vs. Cobenfy



Source: Morgan Stanley Research

Exhibit 33: ML007-C-MA ADP patients treated vs. Cobenfy



Source: Morgan Stanley Research

We model \$600M in risk-adjusted revenues for ADP (vs. \$900M in schizophrenia). We estimate ~3M AD patients with psychosis at 2030 launch year, representing an addressable population of 1.8M (similar to the proportion estimated on off-label antipsychotics). We risk-adjust via a 50% probability of success (vs. 60% in schizophrenia), and forecast \$1.2B in unadjusted revenues or \$590M risk-adjusted (Exhibit 34).

Exhibit 34: Market model for ML-007C-MA for treatment of ADP

Alzheimer's Disease Psychosis (ADP) - US		2025A	2026A	2027A	2028A	2029A	2030A	2031A	2032A	2033A	2034A	2035A	2036A
US population age 65 or older (M)	2%	59.2	60.1	61.0	61.9	62.9	63.8	64.8	65.7	66.7	67.7	68.7	69.8
Alzheimer's disease prevalence (M)	11%	6.6	6.7	6.8	6.9	7.0	7.1	7.2	7.3	7.4	7.5	7.6	7.8
Psychotic symptoms (M)	40%	2.6	2.7	2.7	2.8	2.8	2.8	2.9	2.9	3.0	3.0	3.1	3.1
Patients seeking treatment	60%	1.6	1.6	1.6	1.7	1.7	1.7	1.7	1.8	1.8	1.8	1.8	1.9
Market share							0.5%	1.3%	2.0%	2.8%	3.5%	4.3%	5.0%
Patients treated							8,507	21,586	35,056	48,925	63,202	77,896	93,017
Annual gross price (\$)							24,720	25,462	26,225	27,012	27,823	28,657	29,517
% Y/Y							3%	3%	3%	3%	3%	3%	3%
Annual net price (\$)							18,416	18,842	19,276	19,719	20,171	20,633	21,105
Gross to net							26%	26%	27%	27%	28%	28%	29%
Compliance	60%												
ML-007c unadjusted US sales (\$M)							\$94	\$244	\$405	\$579	\$765	\$964	\$1,178
ML-007c risk-adj. US sales (\$M)	50%						47	122	203	289	382	482	589

Source: Morgan Stanley Research

Debate 3: Can ML004 help address the growing burden of Autism?

Overview: ML004 is a 5-HT1B/1D receptor agonist being evaluated as part of the Phase 2 IRIS study for treatment of core symptoms in Autism Spectrum Disorder (ASD). There are no approved drugs for core autism symptom, however two antipsychotics are approved for treating irritability in ASD, risperidone and aripiprazole.

Street's take: While schizophrenia and use of muscarinics is largely de-risked, ASD has limited benchmarks to evaluate clinical potential or the commercial opportunity. That said, the disorder is highly prevalent. Regulators are pointing to the scientific merit and clinical association behind [cerebral folate deficiency](#) as a driving pathology, and Welcovirin speaks to the regulatory urgency for novel effective therapies following 5x increase in autism over the last two decades.

Our take: We note there are no approved drugs for core autism symptoms (social communication, repetitive behaviors), and less is understood on the underlying pathophysiology. Indeed, prevalence of Autism is readily increasing (+500% over the last two decades), so any FDA approval for treating the core symptoms of the disease would represent a first-in-class approval and confer uncontested market share in a highly prevalent and growing CNS disorder.

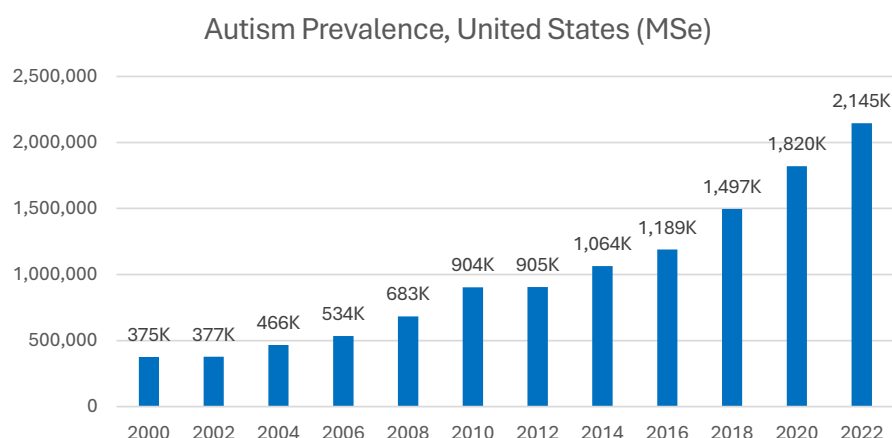
ML004 can address a large unmet need recognized by regulators. In autism, there is strong regulatory desire to fast-track effective medications. To date, risperidone and aripiprazole are the only FDA-approved antipsychotics in ASD (for the treatment of irritability), in stark contrast to schizophrenia (where there are dozens of atypical antipsychotics for schizophrenia). ML004 Phase 2 IRIS is ongoing. Considering precedent set by risperidone and aripiprazole, plus potential for similar pharmacology to 5-HT1B/1D agonist ML-004, we believe ML004 is being largely overlooked and can become a value driver in the not too distant future.

Prevalence of Autism demonstrates that it is a growing national concern, increasing +500% over the last two decades. For comparison, in the early 2000s schizophrenia was estimated in [5.1 per 1,000 lives](#) (~1 in 200), and while the population has grown, schizophrenia remains estimated in [~1 in 200](#). Indeed, the prevalence rate of schizophrenia has remained largely stable over the last two decades, whereas for Autism, data from cdc.gov across ADDM Network Sites demonstrate a ~5x or 500% increase in Autism prevalence ([Exhibit 35](#), [Exhibit 36](#)), increasing from 1:150 (2000) to 1:30 (2022).

Exhibit 35: United States is experiencing a 500% increase in rates of Autism in children over the last 20 years

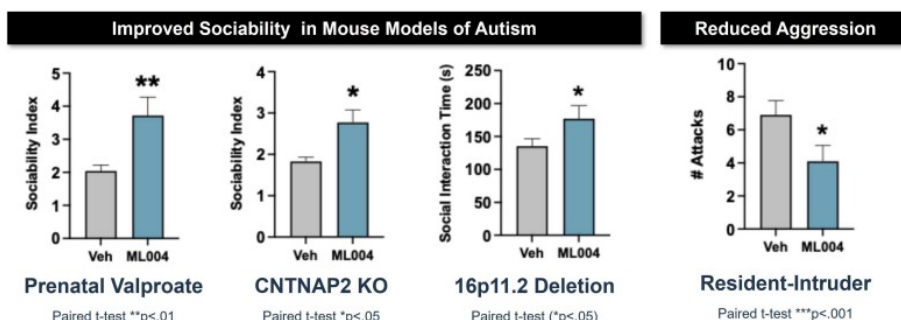
Surveillance Year	Birth Year	Number of ADDM Sites Reporting	Combined Prevalence (per 1,000 Children)	Range Across ADDM Sites	Prevalence	US population (MSe)	US Children Population (Mse, 20%)	ASD population (MSe)
2000	1992	6	6.7	4.5-9.9	1 in 150	280,144,029	56,028,806	375,393
2002	1994	14	6.6	3.3-10.6	1 in 150	285,875,592	57,175,118	377,356
2004	1996	8	8	4.6-9.8	1 in 125	290,999,527	58,199,905	465,599
2006	1998	11	9	4.2-12.1	1 in 110	296,484,368	59,296,874	533,672
2008	2000	14	11.3	4.8-21.2	1 in 88	302,196,033	60,439,207	682,963
2010	2002	11	14.7	5.7-21.9	1 in 68	307,560,345	61,512,069	904,227
2012	2004	11	14.5	8.2-24.6	1 in 69	312,203,803	62,440,761	905,391
2014	2006	11	16.8	13.1-29.3	1 in 59	316,666,589	63,333,318	1,064,000
2016	2008	11	18.5	18.0-19.1	1 in 54	321,367,785	64,273,557	1,189,061
2018	2010	11	23	16.5-38.9	1 in 44	325,475,874	65,095,175	1,497,189
2020	2012	11	27.6	23.1-44.9	1 in 36	329,793,543	65,958,709	1,820,460
2022	2014	16	32.2	9.7 - 53.1	1 in 31	333,103,368	66,620,674	2,145,186

Source: Morgan Stanley Research, www.cdc.gov

Exhibit 36: United States is experiencing a 500% increase in rates of Autism in children over the last 20 years

Source: Morgan Stanley Research, www.cdc.gov

What is ML004, and how can it help to treat core symptoms in ASD? ML-004 is a 5-HT1B/1D receptor agonist, formulated as an immediate-release (IR) and extended-release (ER) oral tablet (based on zolmitriptan chemistry). Preclinical data with optogenetic stimulation of the dorsal raphe -> nucleus accumbens pathway was shown to improved sociability in rodent Autism Spectrum Disorder (ASD) models, like prenatal valproate, CNTNAP2 knock-out, and 16p11.2 deletion ([Exhibit 36](#)). Moreover, blockade of 5-HT1B signaling reversed these effects, supporting ML-004's mechanism of action.

Exhibit 37: ML-004 preclinical results in Autism Spectrum Disorder-model mice

Source: Company reports

Phase 2 IRIS study design. ML004 is being evaluated in Phase 2 IRIS for evaluating improvements in social communication deficits in individuals with ASD, with the goal of achieving a target maintenance dose of 48mg and 72mg ([Exhibit 38](#)).

Exhibit 38: Phase 2 IRIS study summary

Category	Details
Design	Multi-center, randomized, double-blind, placebo-controlled
Sample Size	~150 subjects with ASD
Population	Adolescents (12–17) and Adults (18–45)
Dosing	Flexible dosing; target maintenance doses: 48 mg and 72 mg
Primary Endpoint	ABI Social Communication Domain Score – change from baseline after 12 weeks
Secondary Endpoints	- Change in ABC-I score for patients with moderate+ irritability at baseline - Additional secondary and exploratory measures
Extension	Eligible for 52-week open-label extension (OLE) after Phase 2 completion

Source: Morgan Stanley Research, Company reports

Model Assumptions

We model <\$250M in risk adjusted revenues for ML004 for ASD. We estimate ~2.5M ASD patient at 2031 launch year and a \$20,000 annual list price. We forecast \$780M in peak unadjusted revenues, discounted via a 30% probability of success (Exhibit 39).

Exhibit 39: Market model for ML004 in Autism Spectrum Disorder

Autism Spectrum Disorder (ASD) - US		2022A	2023A	2024A	2025A	2026A	2027A	2028A	2029A	2030A	2031A	2032A	2033A	2034A	2035A	2036A
US adult population (M)	0.8%	260.8	262.9	265.0	267.1	269.3	271.4	273.6	275.8	278.0	280.2	282.5	284.7	287.0	289.3	291.6
ASD prevalence (adults, M)	2.2%	5.8	5.8	5.9	5.9	6.0	6.0	6.1	6.1	6.2	6.2	6.3	6.3	6.4	6.4	6.5
US adolescent population (M)	0.1%	72.5	72.5	72.5	72.6	72.6	72.6	72.7	72.7	72.7	72.8	72.8	72.9	72.9	72.9	73.0
ASD prevalence (adolescents, M)	2.5%	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
ASD prevalence		7.6	7.7	7.7	7.8	7.8	7.8	7.9	7.9	8.0	8.0	8.1	8.1	8.2	8.3	8.3
ASD patients seeking medication treatment (M)	30%	2.3	2.3	2.3	2.3	2.3	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.5	2.5	2.5
Market share										0.5%	1.0%	1.5%	2.0%	2.5%	3.0%	
Patients treated										12,070	24,292	36,669	49,201	61,891	74,740	
Annual gross price (\$)										20,000	20,600	21,218	21,855	22,510	23,185	
%Y/Y											3%	3%	3%	3%	3%	3%
Annual net price (\$)										16,000	16,274	16,550	16,828	17,108	17,389	
Gross to net										20%	21%	22%	23%	24%	25%	
Compliance	60%															
ML-004 unadjusted US sales (\$M)										\$116	\$237	\$364	\$497	\$635	\$780	
ML-004 risk-adj. US sales (\$M)	30%									35	71	109	149	191	234	

Source: Morgan Stanley Research

Financials

Exhibit 40: MapLight Income Statement

(\$M except per share)	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
ML007 Schizophrenia				-	-	-	-	70.4	163.4	262.3	367.2	478.6	596.6	721.5	853.7
ML007 ADP				-	-	-	-	-	47.0	122.0	202.7	289.4	382.5	482.2	588.9
ML004 ASD				-	-	-	-	-	-	34.8	71.2	109.2	149.0	190.6	233.9
Total Revenues	-	-	-	-	-	-	-	70.4	210.4	419.0	641.1	877.2	1,128.0	1,394.3	1,676.6
COGS	-	-	-	-	-	-	-	9.9	29.5	58.7	89.8	122.8	157.9	195.2	234.7
R&D	27.1	49.7	68.5	143.9	179.9	194.3	291.4	364.2	273.2	232.2	243.8	256.0	268.8	282.2	296.4
SG&A	4.1	7.6	14.4	21.6	24.9	26.1	35.3	56.4	104.4	156.6	203.6	254.5	279.9	307.9	338.7
Total expenses	31.2	57.3	82.9	165.5	204.8	220.4	326.7	430.5	407.0	447.5	537.1	633.3	706.6	785.3	869.8
Operating income (loss)	(31.2)	(57.3)	(82.9)	(165.5)	(204.8)	(220.4)	(326.7)	(360.1)	(196.6)	(28.4)	104.0	244.0	421.4	608.9	806.8
Interest income	0.4	1.1	6.4	7.0	7.7	8.5	9.3	10.2	11.3	12.4	13.6	15.0	16.5	18.1	19.9
Other income / (expenses)	0.8	0.5	(1.0)	1.5	1.7	1.8	2.0	2.2	2.4	2.7	2.9	3.2	3.5	3.9	4.3
Foreign exchange gain	0.0	0.0	-	-	-	-	-	-	-	-	-	-	-	-	-
Other Income / (Expense)	1.2	1.6	5.4	8.5	9.3	10.3	11.3	12.4	13.7	15.0	16.5	18.2	20.0	22.0	24.2
Income (loss) before income taxes	(30.0)	(55.7)	(77.6)	(157.0)	(195.4)	(210.1)	(315.4)	(347.7)	(183.0)	(13.4)	120.5	262.1	441.4	631.0	831.0
Income tax	-	-	-	-	-	-	-	-	-	-	(5.1)	(11.0)	(18.5)	(26.5)	(108.0)
Net income (loss)	(30.0)	(55.7)	(77.6)	(157.0)	(195.4)	(210.1)	(315.4)	(347.7)	(183.0)	(13.4)	115.4	251.1	422.9	604.5	723.0
GAAP EPS	(3.21)	(5.03)	(6.27)	\$ (5.49)	\$ (3.76)	\$ (3.57)	\$ (4.67)	\$ (4.44)	\$ (1.99)	\$ (0.13)	\$ 1.15	\$ 2.48	\$ 4.15	\$ 5.91	\$ 7.03
Basic shares	9.4	11.1	12.4	28.6	52.0	58.8	67.6	78.3	91.9	100.1	100.7	101.3	101.8	102.3	102.8
Diluted shares	9.4	11.1	12.4	28.6	52.0	58.8	67.6	78.3	91.9	100.1	100.7	101.3	101.8	102.3	102.8

Source: Morgan Stanley Research, Company reports

Exhibit 41: MapLight Balance Sheet

(\$M)	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Cash and cash equivalents	27.7	79.8	38.3	338.7	242.5	184.1	103.0	73.3	348.0	315.9	418.3	655.6	1,060.0	1,645.5	2,348.8
Marketable securities	-	-	70.5	70.5	70.5	70.5	70.5	70.5	70.5	70.5	70.5	70.5	70.5	70.5	70.5
Prepaid expenses and other current assets	4.4	3.8	5.8	4.6	5.1	5.6	6.2	3.5	10.5	21.0	32.1	43.9	56.4	69.7	83.8
Accounts receivable	-	-	-	-	-	-	-	2.8	8.4	16.8	25.6	35.1	45.1	55.8	67.1
Inventory	-	-	-	-	-	-	-	2.1	6.3	12.6	19.2	26.3	33.8	41.8	50.3
Total current assets	32.0	83.6	114.6	413.9	318.2	260.3	179.8	152.3	443.8	436.7	565.7	831.4	1,265.9	1,883.4	2,620.6
Property and equipment, net	1.0	1.0	1.2	3.0	5.3	7.6	11.4	16.2	20.1	23.9	28.3	33.2	38.3	43.7	49.4
LT investments	-	-	11.4	11.4	11.4	11.4	11.4	11.4	11.4	11.4	11.4	11.4	11.4	11.4	11.4
Restricted cash	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Other assets	6.1	6.6	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5
Total assets	39.4	91.4	136.9	438.0	344.6	289.0	212.2	189.6	485.0	481.8	615.1	885.7	1,325.4	1,948.1	2,691.1
Accounts payable	3.6	3.3	1.9	5.0	6.1	6.6	9.8	12.6	11.3	11.7	13.4	15.3	16.5	17.7	19.1
Accrued expenses	2.8	6.8	10.0	19.5	24.1	26.0	38.5	49.6	44.5	45.8	52.8	60.2	64.7	69.6	74.9
Other current liabilities	6.2	4.7	4.0	4.4	4.9	5.3	5.9	6.5	7.1	7.8	8.6	9.5	10.4	11.5	12.6
Total current liabilities	12.6	14.7	15.9	28.9	35.1	37.9	54.2	68.7	63.0	65.3	74.8	85.0	91.6	98.8	106.5
Long term debt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other liabilities	5.9	5.4	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8
Total liabilities	18.5	20.1	21.7	34.7	40.9	43.7	60.0	74.5	68.8	71.1	80.6	90.8	97.4	104.6	112.3
Preferred stock	83.8	188.9	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8
Common stock	-	-	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6
Additional paid-in capital	3.1	4.2	0.2	445.3	541.1	692.8	915.1	1,225.8	1,709.8	1,717.6	1,726.1	1,735.3	1,745.5	1,756.6	1,768.8
Accumulated deficit	(66.1)	(121.8)	(199.4)	(356.4)	(551.8)	(761.9)	(1,077.3)	(1,425.0)	(1,608.0)	(1,621.4)	(1,505.9)	(1,254.8)	(831.9)	(227.4)	495.6
Total stockholders' equity	20.9	71.3	115.2	403.3	303.6	245.3	152.2	115.1	416.2	410.6	534.5	794.9	1,228.0	1,843.6	2,578.7
Total liabilities and stockholders' equity	39.4	91.4	136.9	438.0	344.6	289.0	212.2	189.6	485.0	481.8	615.1	885.7	1,325.4	1,948.1	2,691.1

Source: Morgan Stanley Research, Company reports

Exhibit 42: MapLight Cash Flow Statement

(\$M)	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Net income (loss)	(30.0)	(55.7)	(77.6)	(157.0)	(195.4)	(210.1)	(315.4)	(347.7)	(183.0)	(13.4)	115.4	251.1	422.9	604.5	723.0
Depreciation	0.6	0.6	0.7	0.7	0.8	1.0	1.2	1.5	1.8	2.0	2.4	2.7	3.1	3.5	3.8
Stock-based compensation	1.4	1.0	1.1	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.9	3.2	3.5	3.9
Other non-cash items	(0.2)	0.1	0.0	-	-	-	-	-	-	-	-	-	-	-	-
Change in operating assets and liabilities:															
Prepaid expenses and other current assets	(2.1)	0.5	(2.3)	1.2	(0.5)	(0.5)	(0.6)	2.7	(7.0)	(10.4)	(11.1)	(11.8)	(12.5)	(13.3)	(14.1)
Accounts payable	1.3	(0.4)	(1.3)	3.1	1.2	0.5	3.2	2.8	(1.3)	0.3	1.8	1.9	1.1	1.2	1.3
Accrued expenses	1.1	3.9	2.2	9.5	4.6	1.8	12.5	11.1	(5.1)	1.3	6.9	7.4	4.5	4.9	5.3
Accounts receivable	-	-	-	-	-	-	-	(2.8)	(5.6)	(8.3)	(8.9)	(9.4)	(10.0)	(10.6)	(11.3)
Inventory	-	-	-	-	-	-	-	(2.1)	(4.2)	(6.3)	(6.7)	(7.1)	(7.5)	(8.0)	(8.5)
Other	1.4	(2.0)	(1.6)	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.8	0.9	0.9	1.0	1.1
Net cash used in operating activities	(26.6)	(52.0)	(78.8)	(141.0)	(187.4)	(205.2)	(296.7)	(332.0)	(201.5)	(31.6)	103.2	238.6	405.7	586.7	704.6
Purchases of property and equipment	(0.7)	(0.5)	(0.8)	(2.5)	(3.1)	(3.3)	(4.9)	(6.3)	(5.7)	(5.8)	(6.7)	(7.7)	(8.2)	(8.9)	(9.5)
Purchases of investments	-	-	(125.0)	-	-	-	-	-	-	-	-	-	-	-	-
Proceeds from maturities of investments	-	-	45.0	-	-	-	-	-	-	-	-	-	-	-	-
Proceeds from sales of investments	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Proceeds from sale of property and equipment	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Net cash (used in) provided by investing activities	(0.7)	(0.5)	(80.8)	(2.5)	(3.1)	(3.3)	(4.9)	(6.3)	(5.7)	(5.8)	(6.7)	(7.7)	(8.2)	(8.9)	(9.5)
Proceeds from issuance of convertible preferred stock, net of offering costs	33.2	104.5	117.8	-	-	-	-	-	-	-	-	-	-	-	-
Proceeds from issuance of common stock, net of offering costs	-	-	-	440.7	90.9	146.4	216.4	304.2	477.0	-	-	-	-	-	-
Proceeds from exercise of stock options and warrants to purchase common stock	0.0	0.0	0.3	3.1	3.4	3.8	4.1	4.5	4.9	5.3	5.8	6.4	7.0	7.6	8.3
Net change in long-term debt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Net cash provided by financing activities	33.2	104.6	118.1	443.9	94.3	150.1	220.5	308.6	461.9	5.3	5.8	6.4	7.0	7.6	8.3
Net increase (decrease) in cash and cash equivalents	5.9	52.1	(41.5)	300.4	(96.1)	(58.4)	(81.1)	(29.7)	274.7	(32.1)	102.4	237.3	404.4	585.5	703.3
Cash and cash equivalents at beginning of the year	22.0	27.9	80.0	38.5	338.9	242.7	184.3	103.2	73.5	348.2	316.1	418.5	655.8	1,060.2	1,645.7
Cash, cash equivalents, and restricted at end of the year	27.9	80.0	38.5	338.9	242.7	184.3	103.2	73.5	348.2	316.1	418.5	655.8	1,060.2	1,645.7	2,349.0
Less restricted cash	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Cash and cash equivalents at end of the year	27.7	79.8	38.3	338.7	242.5	184.1	103.0	73.3	348.0	315.9	418.3	655.6	1,060.0	1,645.5	2,348.8

Source: Morgan Stanley Research, Company reports

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
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(as of October 31, 2025)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1501	41%	384	46%	26%	703	41%
Equal-weight/Hold	1609	44%	370	44%	23%	788	46%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	568	15%	80	10%	14%	223	13%
Total	3,682		835			1715	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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INDUSTRY COVERAGE: Biotechnology

COMPANY (TICKER)	RATING (AS OF)	PRICE* (11/19/2025)
Judah C. Frommer, CFA		
4D Molecular Therapeutics Inc (FDMT.O)	E (11/06/2025)	\$10.80
Abivax (ABVX.O)	O (07/23/2025)	\$112.95
Apellis Pharmaceuticals Inc (APLS.O)	E (11/21/2024)	\$19.45

Arcellx Inc (ACLX.O)	O (03/07/2024)	\$89.62
Arcutis Biotherapeutics, Inc. (ARQT.O)	O (07/02/2025)	\$27.89
Bicara Therapeutics (BCAX.O)	O (10/08/2024)	\$15.18
Celldex Therapeutics Inc (CLDX.O)	O (03/20/2025)	\$24.92
Compass Pathways PLC (CMPS.O)	O (08/08/2025)	\$5.25
enGene Holdings Inc. (ENGN.O)	O (03/08/2024)	\$6.76
Galapagos NV (GLPG.O)	U (02/13/2025)	\$30.49
Genmab A/S (GMAB.O)	++	\$30.57
Incyte Corp (INCY.O)	E (07/02/2025)	\$101.32
Kymera Therapeutics Inc (KYMR.O)	O (07/02/2025)	\$64.82
Mural Oncology (MURA.O)	E (03/25/2025)	\$2.08
ProKidney Corp (PROK.O)	E (11/10/2022)	\$2.32
PTC Therapeutics (PTCT.O)	O (03/06/2025)	\$75.60
Regenxbio Inc (RGNX.O)	O (11/15/2024)	\$10.93
Trevi Therapeutics Inc (TRVI.O)	O (08/21/2025)	\$11.04
Zenas Biopharma Inc (ZBIO.O)	O (07/02/2025)	\$37.15

Maxwell Skor

Adagene Inc. (ADAG.O)	E (01/30/2025)	\$1.81
Amicus Therapeutics Inc. (FOLD.O)	O (07/16/2025)	\$9.73
Ascendis Pharma A/S (ASND.O)	O (05/05/2025)	\$212.15
Atea Pharmaceuticals Inc (AVIR.O)	E (08/12/2024)	\$3.15
Bicycle Therapeutics Plc (BCYC.O)	E (02/14/2022)	\$6.25
Cidara Therapeutics Inc (CDTX.O)	E (11/16/2025)	\$219.02
Crinetics Pharmaceuticals (CRNX.O)	O (01/16/2024)	\$42.90
Cytokinetix Inc (CYTK.O)	O (02/13/2025)	\$64.79
Insmid Inc (INSM.O)	E (08/12/2025)	\$203.56
Pharvaris N.V. (PHVS.O)	O (08/15/2023)	\$24.54
Sana Biotechnology, Inc (SANA.O)	O (03/01/2021)	\$3.40
Tscan Therapeutics Inc (TCRX.O)	E (11/13/2025)	\$1.03
Ultragenyx Pharmaceutical Inc (RARE.O)	O (03/27/2019)	\$31.86

Michael E Ulz

Aardvark Therapeutics Inc. (AARD.O)	O (03/10/2025)	\$10.46
Akero Therapeutics Inc (AKRO.O)	++	\$54.34
Alnylam Pharmaceuticals Inc (ALNY.O)	E (09/09/2022)	\$460.76
Arrowhead Pharmaceuticals Inc (ARWR.O)	E (09/09/2022)	\$38.97
BioAge Labs Inc (BIOA.O)	U (12/09/2024)	\$8.24
Cabaletta Bio Inc (CABA.O)	O (01/27/2023)	\$2.24
Dyne Therapeutics Inc (DYN.O)	O (04/30/2024)	\$20.60
Fractyl Health Inc (GUTS.O)	O (02/27/2024)	\$1.33
IO Biotech (IOBT.O)	E (09/29/2025)	\$0.75
Ionis Pharmaceuticals Inc (IONS.O)	O (07/30/2025)	\$74.00
Kyverna Therapeutics (KYTX.O)	O (03/04/2024)	\$8.23
Mirum Pharmaceuticals (MIRM.O)	O (11/13/2023)	\$70.44
Rhythm Pharmaceuticals Inc (RYTM.O)	O (12/19/2023)	\$103.15
Rocket Pharmaceuticals Inc (RCKT.O)	E (05/27/2025)	\$2.88
Sarepta Therapeutics Inc (SRPT.O)	E (06/16/2025)	\$16.39
Silence Therapeutics Plc (SLN.O)	O (05/08/2023)	\$6.43
Tenaya Therapeutics Inc (TNYA.O)	O (08/24/2021)	\$1.19
Viking Therapeutics Inc (VKTX.O)	O (06/27/2024)	\$36.04
Vir Biotechnology Inc (VIR.O)	O (01/08/2025)	\$5.44
Zentalis Pharmaceuticals Inc (ZNTL.O)	E (06/18/2024)	\$1.33

Sean Laaman, Ph.D.

ABSCI CORP (ABSI.O)	O (07/03/2024)	\$2.86
Acadia Pharmaceuticals Inc (ACAD.O)	E (08/06/2024)	\$23.10
Alector Inc (ALEC.O)	U (11/26/2024)	\$1.22

argenx SE (ARGX.O)	O (07/02/2025)	\$914.59
Axsome Therapeutics (AXSM.O)	O (04/29/2024)	\$146.77
BeOne Medicines (ONC.O)	O (12/02/2024)	\$353.79
Biomarin Pharmaceutical Inc (BMRN.O)	O (07/02/2025)	\$52.56
Centessa Pharmaceuticals, Inc (CNTA.O)	O (09/19/2024)	\$28.00
CG Oncology (CGON.O)	O (02/19/2024)	\$42.60
Contineum Therapeutics (CTNM.O)	O (04/30/2024)	\$12.43
Cullinan Therapeutics (CGEM.O)	O (03/06/2025)	\$8.45
Denali Therapeutics Inc (DNLI.O)	O (03/06/2025)	\$16.92
Disc Medicine Inc (IRON.O)	O (11/04/2024)	\$90.74
Erasca, Inc. (ERAS.O)	E (08/17/2025)	\$2.76
Exelixis Inc. (EXEL.O)	O (01/26/2025)	\$42.14
Halozyne Therapeutics, Inc (HALO.O)	O (08/05/2025)	\$70.25
Immunocore Holdings Ltd (IMCR.O)	E (12/12/2024)	\$38.65
MapLight Therapeutics (MPLT.O)	O (11/21/2025)	\$14.60
Neurocrine Biosciences Inc (NBIX.O)	O (03/06/2025)	\$138.06
Recursion Pharmaceuticals Inc (RXRX.O)	E (05/22/2023)	\$4.03
Schrodinger Inc. (SDGR.O)	E (11/19/2021)	\$16.85

Terence C Flynn, Ph.D.

Alumis Inc. (ALMS.O)	O (05/30/2025)	\$6.57
Amgen Inc. (AMGN.O)	E (10/16/2023)	\$342.40
Arcus Biosciences Inc. (RCUS.N)	O (10/11/2022)	\$20.22
Arvinas Inc (ARVN.O)	E (04/06/2022)	\$11.53
Biogen Inc (BIIB.O)	E (10/31/2024)	\$167.93
Biohaven Ltd (BHAVN.N)	O (07/24/2024)	\$9.46
BioNTech SE (BNTX.O)	O (09/23/2024)	\$101.43
CRISPR Therapeutics AG (CRSP.O)	U (10/11/2022)	\$50.73
Gilead Sciences Inc. (GILD.O)	O (01/10/2025)	\$128.07
Intellia Therapeutics Inc (NTLA.O)	E (01/26/2025)	\$8.24
Legend Biotech Corp (LEGN.O)	O (01/31/2022)	\$28.00
Moderna Inc (MRNA.O)	E (12/16/2020)	\$24.18
Nurix Therapeutics Inc. (NRIX.O)	E (10/11/2022)	\$14.33
Prime Medicine Inc (PRME.O)	E (11/14/2022)	\$3.72
Regeneron Pharmaceuticals Inc. (REGN.O)	O (09/09/2022)	\$702.75
Structure Therapeutics Inc (GPCR.O)	O (09/22/2024)	\$36.51
United Therapeutics Corp (UTHR.O)	E (07/11/2024)	\$476.54
Vertex Pharmaceuticals (VRTX.O)	E (05/03/2022)	\$421.03

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